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(54) **CACAO PLANT NAMED 'ISC-1'**
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ABSTRACT
(57) A new and distinct variety of *cacao* plant named 'ISC-1', particularly selected for its self-compatibility, is disclosed.

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a Divisional of application Ser. No. 18/991,045, filed Dec. 20, 2024, which claims the benefit of U.S. Provisional Application No. 63/614,442, filed Dec. 22, 2023, each of which is hereby incorporated by reference in its entirety.

[0002] Latin name: Botanical classification: *Theobroma cacao* L.

[0003] Varietal denomination: Variety denomination: the varietal denomination of the *cacao* variety disclosed herein is 'ISC-1'.

BACKGROUND OF THE INVENTION

[0004] *Cacao* beans are produced by *cacao* trees, which are found in warm, moist climates in areas about 20 degrees latitude north and south of the Equator. The *cacao* tree produces leaves, flowers and fruit throughout the year, and the ripe fruit or pod resembles a long cantaloupe, typically containing from about 20 to 40 almond-shaped *cacao* beans. *Cacao* butter and *cacao* solids are extracted from commercial *cacao* beans and processed to produce various *cacao* and chocolate products. Different varieties of *cacao* trees have varying *cacao* bean yield, resistance to disease, and levels of compounds that impact the flavor and nutritional value of *cacao* products. *Cacao* is an important crop that has been appreciated by humans for thousands of years.

[0005] *Theobroma cacao* L. trees predominantly outcross, with most trees being considered self-incompatible (also referred to as "SI"), meaning that other compatible clones must be planted with them to facilitate successful pollination and seed set. In some growing areas (e.g., Central America) many cultivated trees have undergone human-mediated selection for self-compatibility (also referred to as "SC"). However, in other growing areas (e.g., Indonesia), all of the *cacao* clones grown are self-incompatible, and the resulting need for other compatible clones to be planted with them is often not recognized by farmers. Monoclonal plantings are extremely common and severely curtail yields. There is a need for new self-compatible *cacao* clones to remove the risk of monoclonal plantings. Self-compatible *cacao* clones ensure that all pollination events have the potential to be successful and ensure high yields.

SUMMARY OF THE INVENTION

[0006] In order to meet these needs, the present invention is directed to an improved variety of *cacao* plant. In particular, the invention relates to a new and distinct variety of *cacao* plant (*Theobroma cacao* L.) that has been denominated as 'ISC-1'. Variety 'ISC-1' is a self-compatible clone and is high yielding (more than 2.5 t/ha) and suitable for monoclonal plantations. When compared to the industry standard non-selfing clone 'MCC02', cumulative yields at year three of production of the self-compatible *cacao* clone of the present disclosure had a 3 t DB/ha advantage. Further, the disease tolerance of the self-compatible *cacao* clone appears to be good.

[0007] The new *cacao* plant of the present invention is stable and reproduces true to type in successive generations of asexual reproduction.

[0008] *Cacao* plant variety 'ISC-1' originated from a controlled cross between the *cacao* female parent 'MCC02' (unpatented) and the *cacao* male parent 'RUQ1347' (unpatented). 'ISC-1' was asexually propagated via grafting onto rootstock in Pangkep, Indonesia. 'ISC-1' was particularly selected for its self-compatibility, in addition to the plant having high yield potential.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] This new *cacao* plant is illustrated by the accompanying photographs. The colors shown are as true as can be reasonably obtained by conventional photographic procedures.

[0010] FIGS. 1A-1F illustrate leaves of variety 'ISC-1'. FIG. 1A illustrates the length of a leaf blade of variety 'ISC-1'. FIG. 1B illustrates the width of a leaf blade of variety 'ISC-1'. FIG. 1C illustrates the shape of base of a leaf blade of variety 'ISC-1'. FIG. 1D illustrates the intensity of green color of a leaf blade of variety 'ISC-1'. FIG. 1E illustrates the shape of apex of a leaf blade of variety 'ISC-1'. FIG. 1F illustrates the colors of young leaves of variety 'ISC-1' at the stages soon, early, and medium.

[0011] FIGS. 2A-2I illustrate flowers of variety 'ISC-1'. FIG. 2A illustrates the anthocyanin coloration of pedicel of a flower of variety 'ISC-1'. FIG. 2B illustrates the anthocyanin coloration of pedicel of a young flower of variety 'ISC-1'. FIG. 2C illustrates the anthocyanin coloration of

pedicel of a closed flower of variety 'ISC-1'. FIG. 2D illustrates the anthocyanin coloration of pedicel of an open flower of variety 'ISC-1'. FIG. 2E illustrates the length of sepal of a flower of variety 'ISC-1'. FIG. 2F illustrates the width of sepal of a flower of variety 'ISC-1'. FIG. 2G illustrates the anthocyanin coloration of sepal of a flower of variety 'ISC-1'. FIG. 2H illustrates the color of ligula of a flower of variety 'ISC-1'. FIG. 2I illustrates the anthocyanin coloration of staminode of a flower of variety 'ISC-1'.

[0012] FIGS. 3A-3J illustrate fruit of variety 'ISC-1'. FIG. 3A illustrates the shape of a fruit of variety 'ISC-1'. FIG. 3B illustrates the basal constriction of a fruit of variety 'ISC-1'. FIG. 3C illustrates the shape of apex of a fruit of variety 'ISC-1'. FIG. 3D illustrates the fruit length of variety 'ISC-1'. FIG. 3E illustrates the fruit diameter of variety 'ISC-1'. FIG. 3F illustrates the fruit surface of variety 'ISC-1'. FIG. 3G illustrates the fruit depth of grooves of variety 'ISC-1'. FIG. 3H illustrates the fruit color of variety 'ISC-1'. FIG. 3I illustrates the exocarp thickness of a fruit of variety 'ISC-1'. FIG. 3J illustrates the color of pulp of a fruit of variety 'ISC-1'.

[0013] FIGS. 4A-4F illustrate seeds of variety 'ISC-1'. FIG. 4A illustrates the shape of a seed of variety 'ISC-1'. FIG. 4B illustrates the shape in longitudinal section of a seed of variety 'ISC-1'. FIG. 4C illustrates the seed length of variety 'ISC-1'. FIG. 4D illustrates the seed width of variety 'ISC-1'. FIG. 4E illustrates the seed thickness of variety 'ISC-1'. FIG. 4F illustrates the seed cotyledon color of variety 'ISC-1'.

[0014] FIGS. 5A-5F illustrate leaves of comparison variety 'MCC02'. FIG. 5A illustrates the length of a leaf blade of comparison variety 'MCC02'. FIG. 5B illustrates the width of a leaf blade of comparison variety 'MCC02'. FIG. 5C illustrates the shape of base of a leaf blade of comparison variety 'MCC02'. FIG. 5D illustrates the intensity of green color of a leaf blade of comparison variety 'MCC02'. FIG. 5E illustrates the shape of apex of a leaf blade of comparison variety 'MCC02'. FIG. 5F illustrates the colors of young leaves of comparison variety 'MCC02' at the stages soon, early, and medium.

[0015] FIGS. 6A-6G illustrate flowers of comparison variety 'MCC02'. FIG. 6A illustrates the anthocyanin coloration of pedicel of a flower of comparison variety 'MCC02'. FIG. 6B illustrates the anthocyanin coloration of pedicel of a flower of comparison variety 'MCC02'. FIG. 6C illustrates the length of sepal of a flower of comparison variety 'MCC02'. FIG. 6D illustrates the width of sepal of a flower of comparison variety 'MCC02'. FIG. 6E illustrates the anthocyanin coloration of sepal of a flower of comparison variety 'MCC02'. FIG. 6F illustrates the color of ligula of a flower of comparison variety 'MCC02'. FIG. 6G illustrates the anthocyanin coloration of staminode of a flower of comparison variety 'MCC02'.

[0016] FIGS. 7A-7J illustrate fruit of comparison variety 'MCC02'. FIG. 7A illustrates the shape of a fruit of comparison variety 'MCC02'. FIG. 7B illustrates the basal constriction of a fruit of comparison variety 'MCC02'. FIG. 7C illustrates the shape of apex of a fruit of comparison variety 'MCC02'. FIG. 7D illustrates the fruit length of comparison variety 'MCC02'. FIG. 7E illustrates the fruit diameter of comparison variety 'MCC02'. FIG. 7F illustrates the fruit surface of comparison variety 'MCC02'. FIG. 7G illustrates the fruit depth of grooves of comparison variety 'MCC02'. FIG. 7H illustrates the fruit color of

comparison variety 'MCC02'. FIG. 7I illustrates the exocarp thickness of a fruit of comparison variety 'MCC02'. FIG. 7J illustrates the color of pulp of a fruit of comparison variety 'MCC02'.

[0017] FIGS. 8A-8E illustrate seeds of comparison variety 'MCC02'. FIG. 8A illustrates the shape in longitudinal section of a seed of comparison variety 'MCC02'. FIG. 8B illustrates the seed length of comparison variety 'MCC02'. FIG. 8C illustrates the seed width of comparison variety 'MCC02'. FIG. 8D illustrates the seed thickness of comparison variety 'MCC02'. FIG. 8E illustrates the seed cotyledon color of comparison variety 'MCC02'.

[0018] FIGS. 9A-9G illustrate the superior yield of the claimed self-compatible *cacao* clone across crop years. FIGS. 9A, C, and F illustrate cumulative yields for select self-compatible *cacao* clones across crop years, wherein each colored series of data points represents a different *cacao* lineage. Self-compatible lineages are denoted by labeled arrows, and comparative commercial variety 'MCC02' is illustrated in orange (FIG. 9A) or black (FIGS. 9C, and F). FIGS. 9B, D, E, and G illustrate yield per year for many individual clones for each of two (FIG. 9G) or three (FIGS. 9B, D, and E) crop years, wherein the measured clones are illustrated by orange bars, and the green bar illustrates the yield of comparative commercial variety 'MCC02'. FIGS. 9A-9B illustrate yield for Clone Trial 95 ("CT95"), measuring 1,500 trees/Ha planted in May 2018. FIGS. 9C-9D further illustrate CT95, measuring 2,000 trees/Ha. FIG. 9E illustrates yield for Clone Trial 107 ("CT107") planted in August 2018. FIGS. 9F-9G illustrate yield for Clone Trial 162 ("CT162") planted in May 2019. Variety 'ISC-1' is represented in FIG. 9G by the clones labeled "2" (circled in red).

[0019] FIGS. 10A-10C illustrate additional data demonstrating the superior yield of the claimed self-compatible *cacao* clone across crop years. FIG. 10A further illustrates cumulative yield (in Ton/Ha) from CT95 at 1,500 trees/Ha over crop age 2-6 years, with reference variety 'MCC02' represented by the clone labeled "MCC02" (dark purple). FIG. 10B further illustrates cumulative yield (in Ton/Ha) from CT162 over crop age 2-5 years, with variety 'ISC-1' represented by the clone labeled "2IGTB04C09R19|MCC02 x RUQ 1347" (green), and reference variety 'MCC02' represented by the clone labeled "MCC02" (dark purple). FIG. 10C further illustrates cumulative yield (in Ton/Ha) from CT95 at 2,000 trees/Ha over crop age 2-6 years, with reference variety 'MCC02' represented by the clone labeled "MCC02" (dark purple).

[0020] FIG. 11 illustrates rates of survival for self-pollinated individuals during progress testing of variety 'ISC-1'. "REP" represents replicate experiments 1-6 in the leftmost column. The number of flowers initially pollinated is shown in the column second from left. The number of surviving flowers for each replicate are displayed in columns from left to right, tracking the number of survivors at 3-, 5-, 8-, 15-, and 30-days post-pollination, respectively. The final rate of survival is shown in the second-from-right column, calculated as the number of survivors at 30-days post-pollination divided by the number of flowers initially pollinated. The grand survival rate for each column is shown as a percentage across the bottom. The average survival rates are shown in the rightmost column. The standard deviation (StDev) is shown in the bottom right corner.

DETAILED BOTANICAL DESCRIPTION

[0021] The following description is based on observations taken of plants and fruits of the new and distinctive *cacao* variety of the present invention.

[0022] The dry weight of a single bean is calculated by dividing the dry bean weight by the number of healthy beans harvested. The yield is calculated by dividing the total fresh bean weight by the number of reps and multiplying by the plant density (1,175 trees/ha, unless indicated otherwise).

Variety 'ISC-1'

[0023] In certain aspects, the present invention relates to a new and distinctive *cacao* plant designated as 'ISC-1'. *Cacao* variety 'ISC-1' has shown uniformity and stability for the traits, within the limits of environmental influence for the traits. *Cacao* variety 'ISC-1' has been increased with continued observation for uniformity. No variant traits have been observed or are expected in 'ISC-1'.

[0024] The variety description information in Table 1 below provides a summary of variety 'ISC-1' plant characteristics. Those of skill in the art will recognize that these are typical values that may vary due to environment, and that other values that are substantially equivalent are within the scope of the invention. The terminology and descriptors are in line with the descriptors of the "UPOV Guidelines for the Conduct of Tests for Distinctness, Uniformity, and Stability", or the "Test Guidelines" for *Theobroma cacao* L. The "Test Guidelines" indicate reference varieties for the descriptors or characteristics that are included in the list. The terminology and descriptors used in these tables are in line with the official terminology as of the filing date, and are thus clear for a person skilled in the art. FIGS. 1A-4F illustrate the plant characteristics of variety 'ISC-1'. FIGS. 9A-9G illustrate the success of the methods used to produce self-compatible variety 'ISC-1', and FIG. 9G illustrates the yield of variety 'ISC-1' as clone "2". FIG. 10B illustrates additional data demonstrating the superior yield of variety 'ISC-1' across crop years (shown in green and labeled "2|GTB04C09R19|MCC02 x RUQ 1347"). FIG. 11 illustrates the progress testing survival rate data for variety 'ISC-1'.

[0025] The following detailed descriptions set forth the distinctive characteristics of 'ISC-1'. The data which define these characteristics is based on observations taken in Pangkep, Indonesia. This description is in accordance with UPOV terminology. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic, and cultural conditions. 'ISC-1' has not been observed under all possible environmental conditions. The indicated values represent averages calculated from measurements of several plants. Color references are primarily to The RHS Colour Chart of The Royal Horticultural Society of London (R.H.S.) (2015 edition). Descriptive terminology follows the Plant Identification Terminology, An Illustrated Glossary, 2nd edition by James G. Harris and Melinda Woolf Harris, unless where otherwise defined.

[0026] The *cacao* variety 'ISC-1' originated from a controlled cross between varieties 'MCC02' and 'RUQ1347'. The breeder's reference name of variety 'ISC-1' is 'GTB04C09R19'. *Cacao* variety 'ISC-1' has subsequently been asexually propagated via grafting onto rootstock.

[0027] Classification:

[0028] *Species*.—*Theobroma cacao* L.

[0029] *Common name*.—*Cacao*.

[0030] *Denomination*.—'ISC-1'.

[0031] Parentage:

[0032] *Female parent*.—'MCC02' (unpatented).

[0033] *Male parent*.—'RUQ1347' (unpatented).

TABLE 1

'ISC-1'.	
Characteristic	Value
Leaf blade: length	30 cm
Leaf blade: width	10.7 cm
Leaf blade: shape of base	Obtuse
Leaf blade: intensity of green color	Dark
Leaf blade: apex	Acuminate
Young leaf: color - soon	Brown-Dark red
Young leaf: color - early	Dark red
Young leaf: color - medium	Medium brown
Flower: anthocyanin coloration of pedicel	Strong-moderate
Flower: length of sepal	0.7 cm
Flower: width of sepal	0.3 cm
Flower: anthocyanin coloration of sepal	Moderate
Flower: color of ligula	Yellow
Staminode: anthocyanin coloration	Strong
Fruit: shape	Oblong
Fruit: basal constriction	Absent/Very Weak
Fruit: shape of apex	Obtuse
Fruit: length	17.07 cm
Fruit: diameter	10.31 cm
Fruit: length/diameter ratio	1.66:1
Fruit: surface	Moderately rough
Fruit: depth of groove	Deep
Fruit: color	Medium red
Fruit: exocarp thickness	Thick; 2.08 cm
Fruit: color of pulp	White
Fruit: sweetness of pulp	μ19.72° Brix
Fruit: number of seeds	Pod 1: 39 seeds Pod 2: 35 seeds
Seed: shape in longitudinal section	Elliptic
Seed: length	3.04 cm
Seed: width	1.54 cm
Seed: ratio length/width	1.97:1
Seed: thickness	0.89 cm
Seed: cotyledon color	Purple

[0034] The self-compatible phenotype of *cacao* variety 'ISC-1' has been confirmed through field observation by performing controlled (hand) self-pollinations.

COMPARISON WITH PARENTAL AND REFERENCE VARIETY

[0035] *Cacao* variety 'MCC02' is the industry standard non-selfing clone. It may be used as a comparison variety for the self-compatible clone 'ISC-1'.

[0036] The variety description information in Table 2 provides a summary of comparison variety 'MCC02' plant characteristics. Those of skill in the art will recognize that these are typical values that may vary due to environment, and that other values that are substantially equivalent are within the scope of the invention. The terminology and descriptors are in line with the descriptors of the "UPOV Guidelines for the Conduct of Tests for Distinctness, Uniformity, and Stability", or the "Test Guidelines" for *Theobroma cacao* L. The "Test Guidelines" indicate reference varieties for the descriptors or characteristics that are included in the list. The terminology and descriptors used in these tables are in line with the official terminology as of the

filing date and are thus clear for a person skilled in the art. FIGS. 5A-8E illustrate the plant characteristics of comparison variety ‘MCC02’.

TABLE 2

Comparison variety ‘MCC02’	
Characteristic	Value
Leaf blade: length	35.9 cm
Leaf blade: width	11.7 cm
Leaf blade: shape of base	Acute
Leaf blade: intensity of green color	Dark
Leaf blade: apex	Acuminate
Young leaf: color - soon	Brown-medium red
Young leaf: color - early	Light red
Young leaf: color - medium	Medium green
Flower: anthocyanin coloration of pedicel	Strong
Flower: length of sepal	0.7 cm
Flower: width of sepal	0.2 cm
Flower: anthocyanin coloration of sepal	Moderate
Flower: color of ligula	Yellow
Staminode: anthocyanin coloration	Strong
Fruit: shape	Obovate
Fruit: basal constriction	Weak
Fruit: shape of apex	Obtuse
Fruit: length	18 cm
Fruit: diameter	9.92 cm
Fruit: length/diameter ratio	1.81:1
Fruit: surface	Slightly rough
Fruit: depth of grooves	Shallow
Fruit: color	Medium red-orange
Fruit: exocarp thickness	Medium thickness: 1.94
Fruit: color of pulp	White
Fruit: sweetness of pulp	Freshness: μ 17.14° Brix
Fruit: number of seeds	Pod 1: 45 beans Pod 2: 45 beans

TABLE 2-continued

Comparison variety ‘MCC02’	
Characteristic	Value
Seed: shape in longitudinal section	Oblong
Seed: length	3.55 cm
Seed: width	1.66 cm
Seed: ratio length/width	2.14:1
Seed: thickness	1.12 cm
Seed: cotyledon color	Purple

[0037] ‘ISC-1’ differs from the female parent *cacao* plant and reference *cacao* variety ‘MCC02’ (unpatented) in that ‘ISC-1’ has improved yield compared to ‘MCC02’, and in that ‘ISC-1’ is self-compatible while ‘MCC02’ is non-selfing.

REFERENCES

[0038] Glendinning D R. 1967. Incompatibility alleles of cocoa. *Nature* 213, 306.

[0039] Marshall J (1933) Fertility in *cacao*. 3rd Annual Report on *Cacao* Research (Trinidad), p 34

[0040] Pound, F. J. “Studies of fruitfulness in *cacao*. I. *A note on the abscission of the flower*. II. *Evidence for partial sterility*.” *Studies of fruitfulness in cacao*. I. A note on the abscission of the flower. II. Evidence for partial sterility. (1932).

We claim:

1. A new and distinct variety of *cacao* plant named ‘ISC-1’ as shown and described herein.

* * * * *

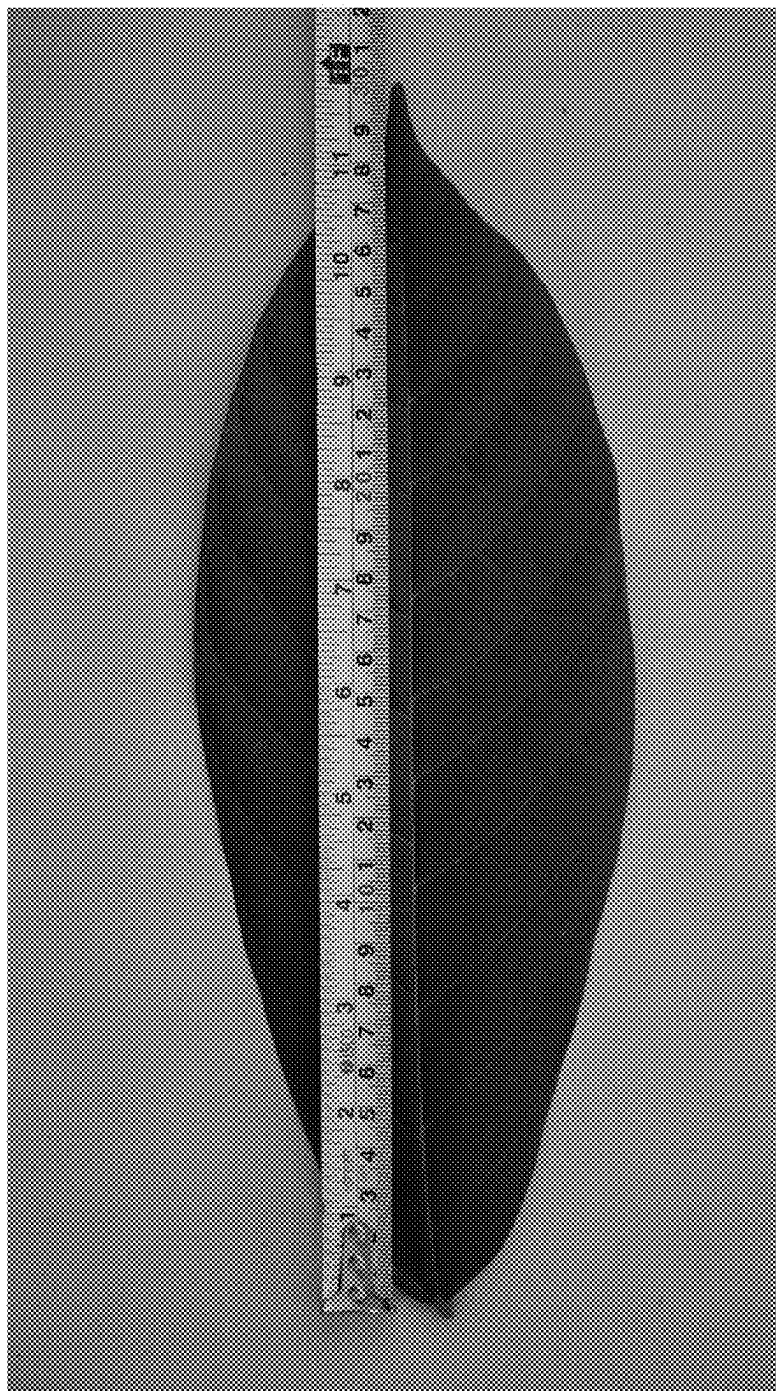


FIG. 1A

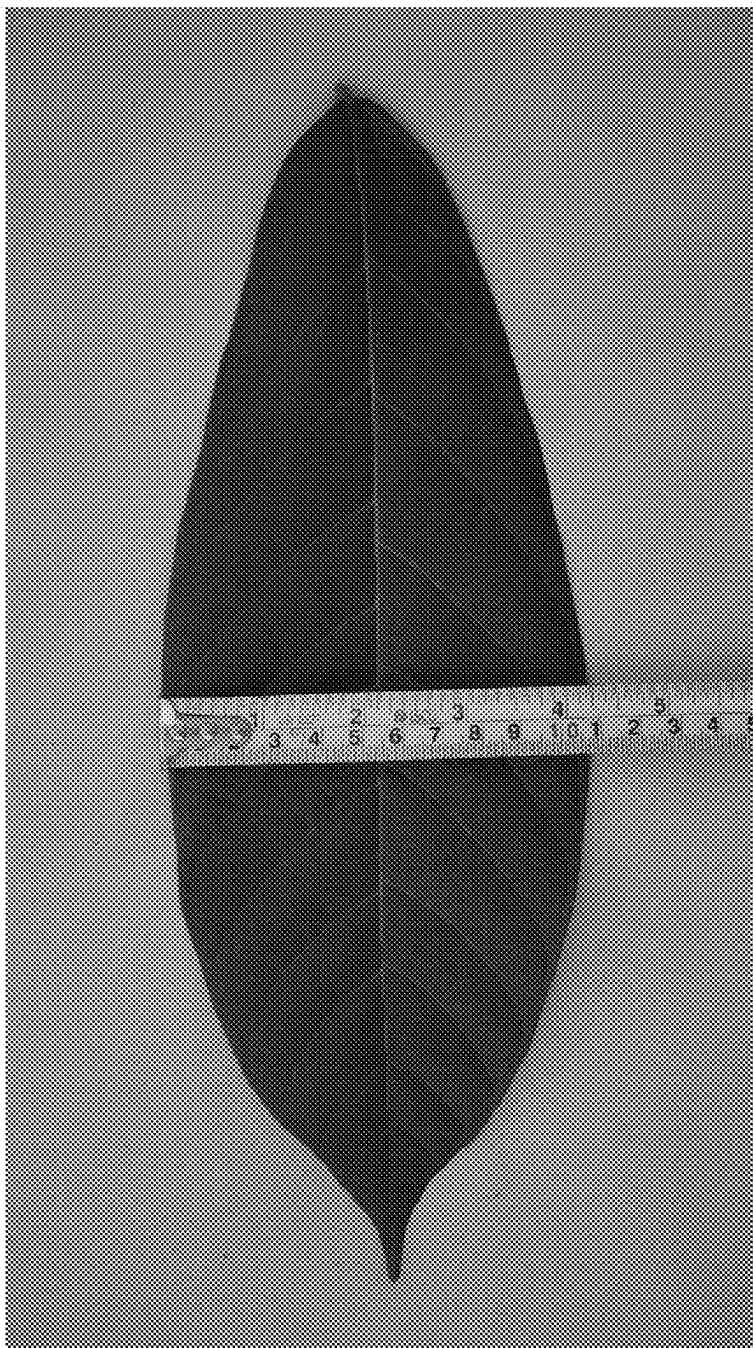


FIG. 1B



FIG. 1C

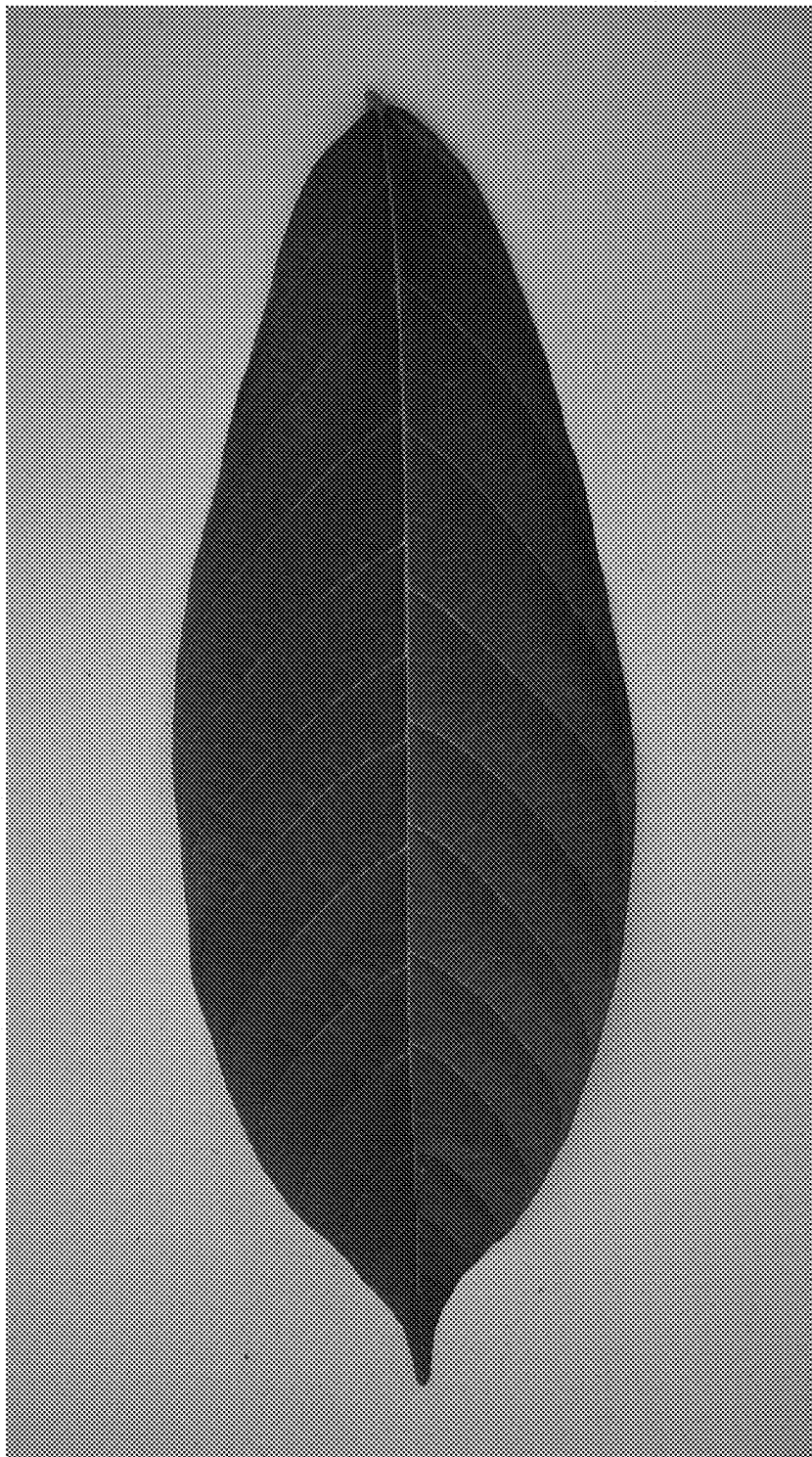


FIG. 1D

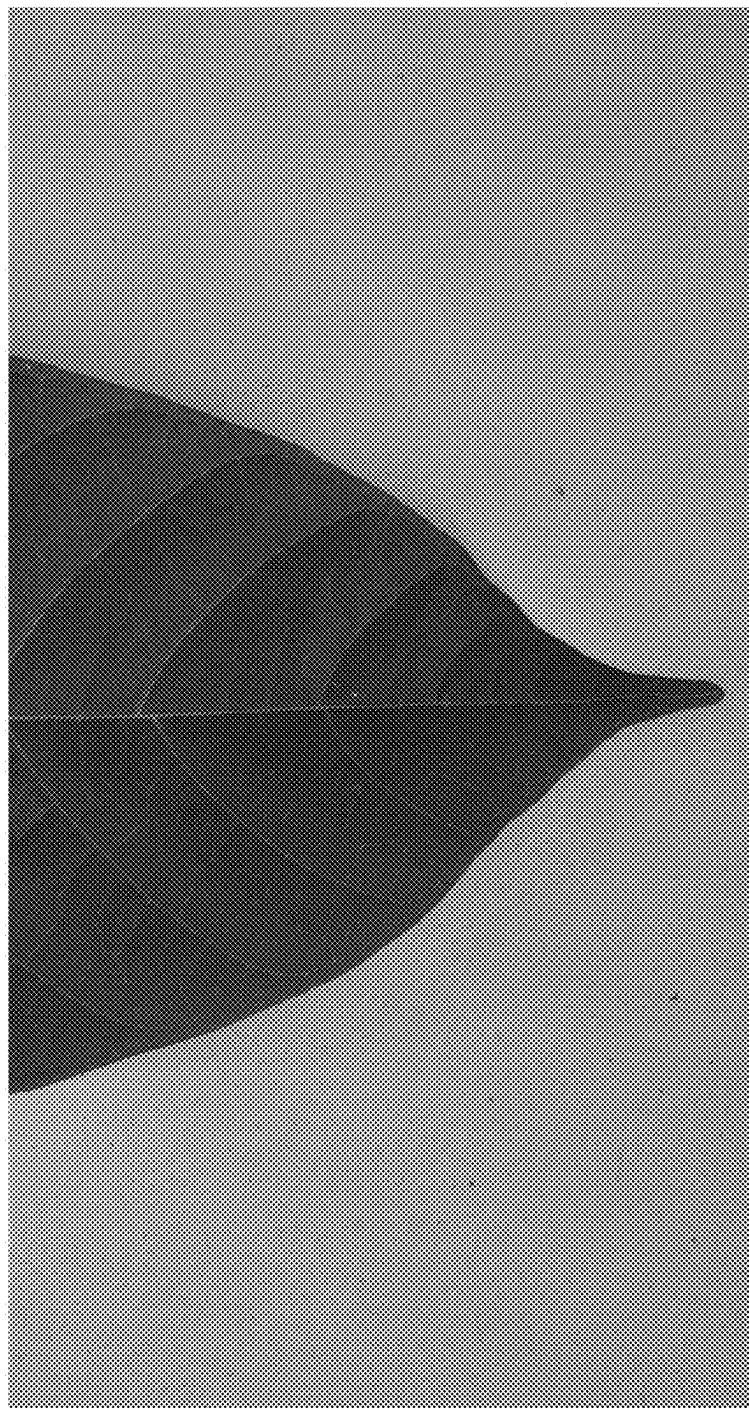


FIG. 1E

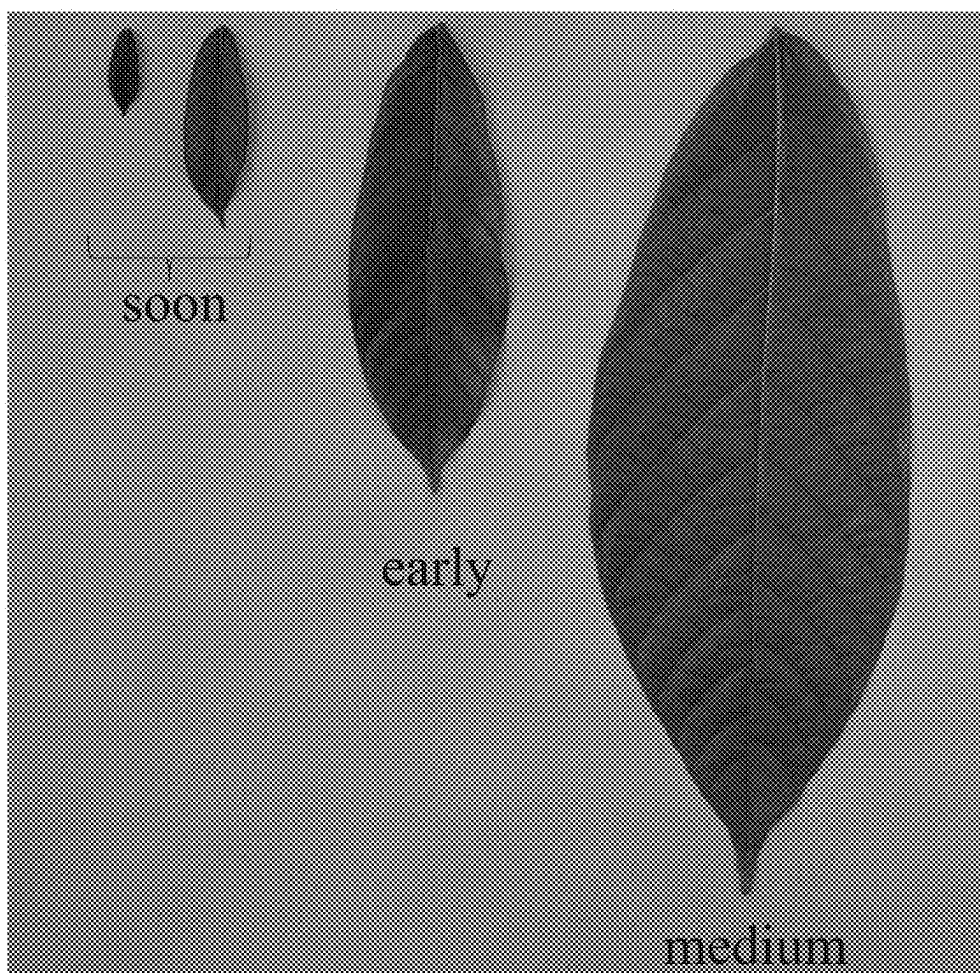


FIG. 1F



FIG. 2A

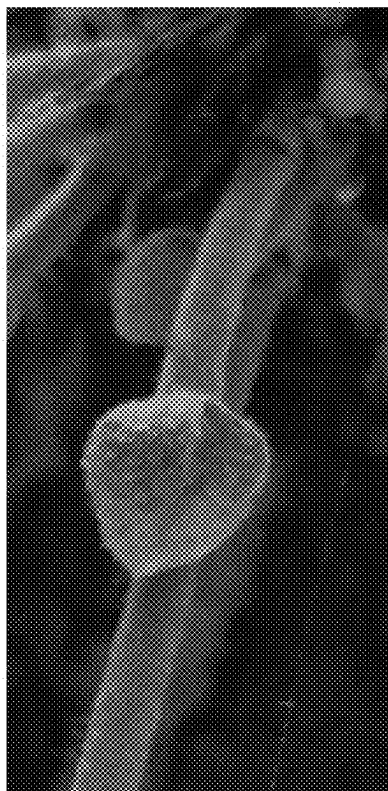


FIG. 2B

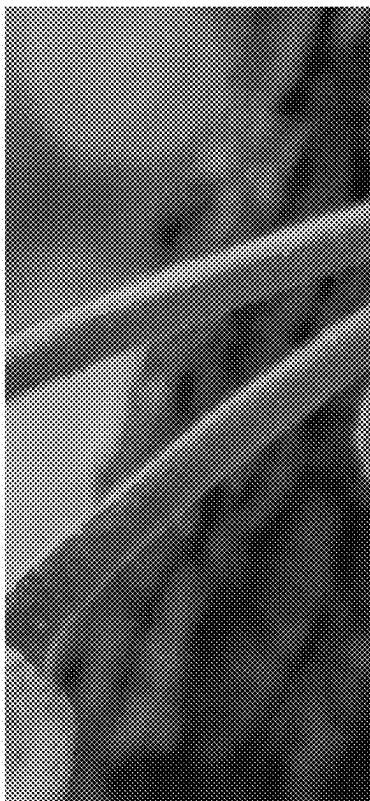


FIG. 2C

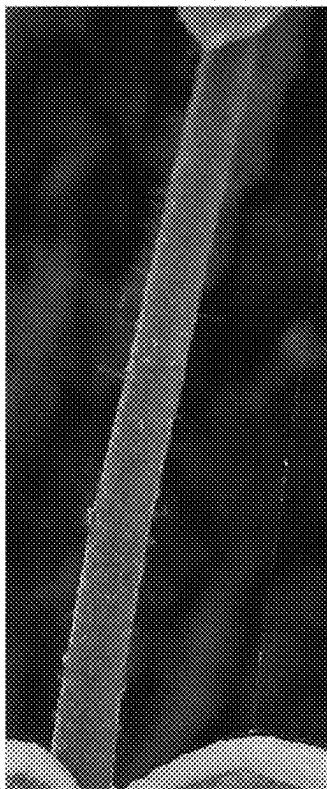


FIG. 2D

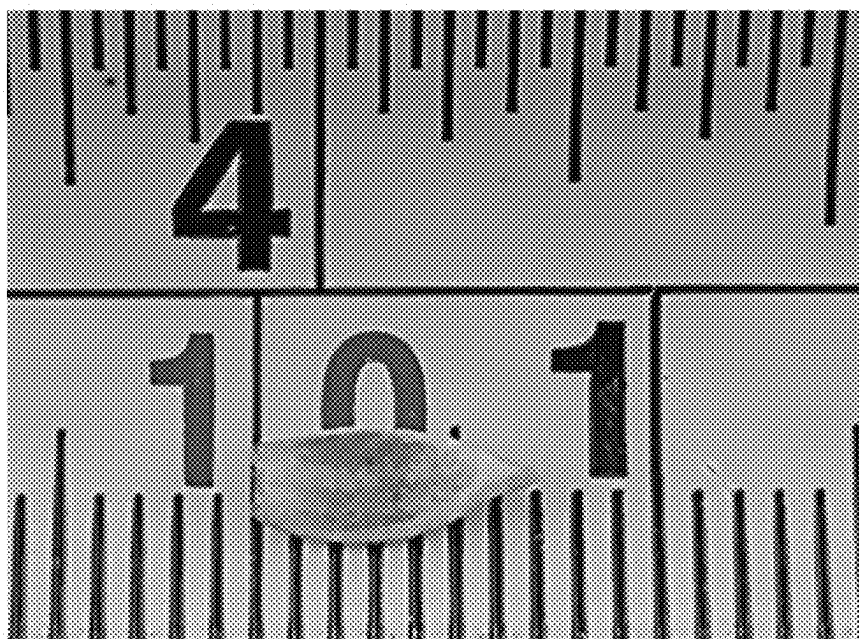


FIG. 2E



FIG. 2F



FIG. 2G



FIG. 2H

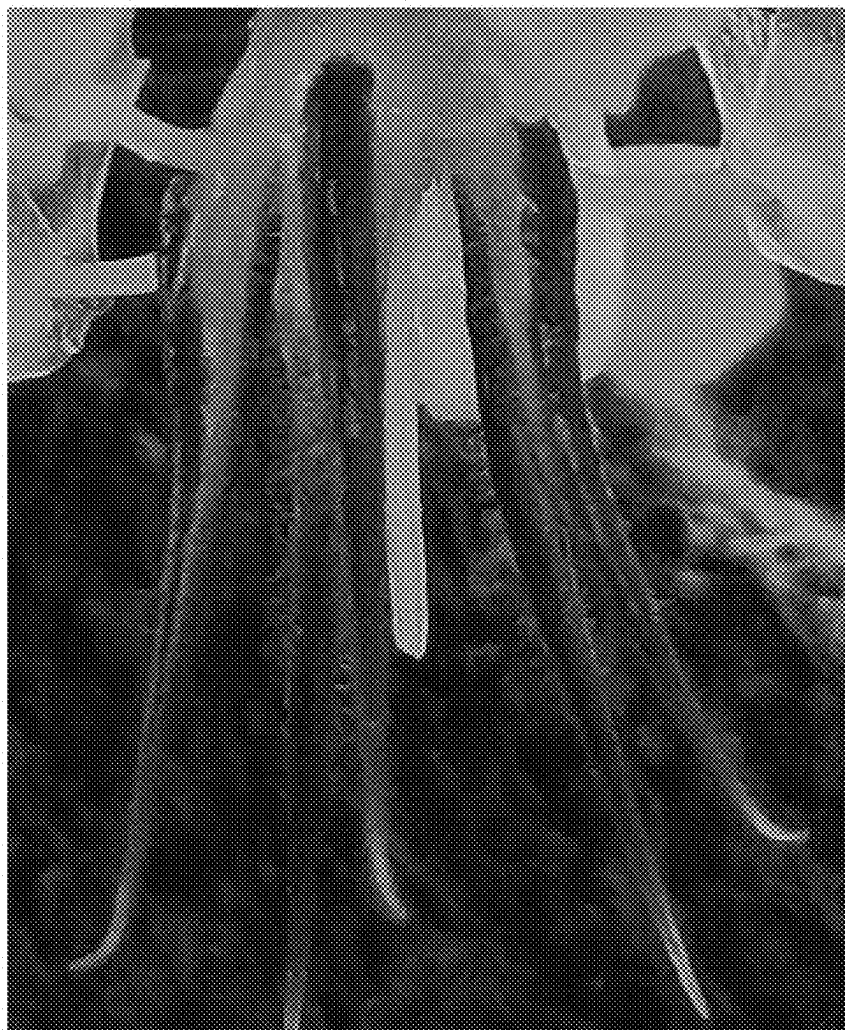


FIG. 2I



FIG. 3A

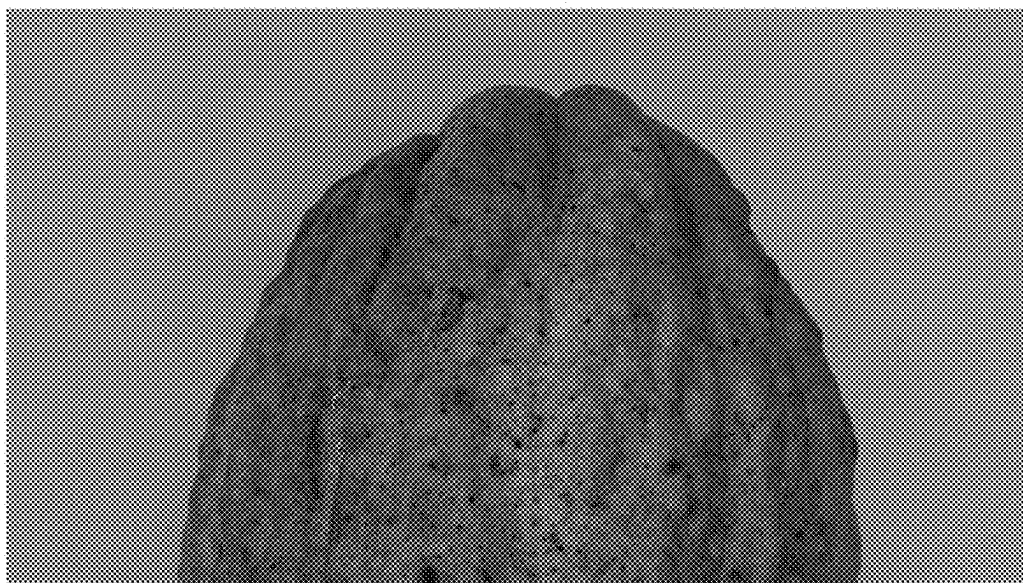


FIG. 3B

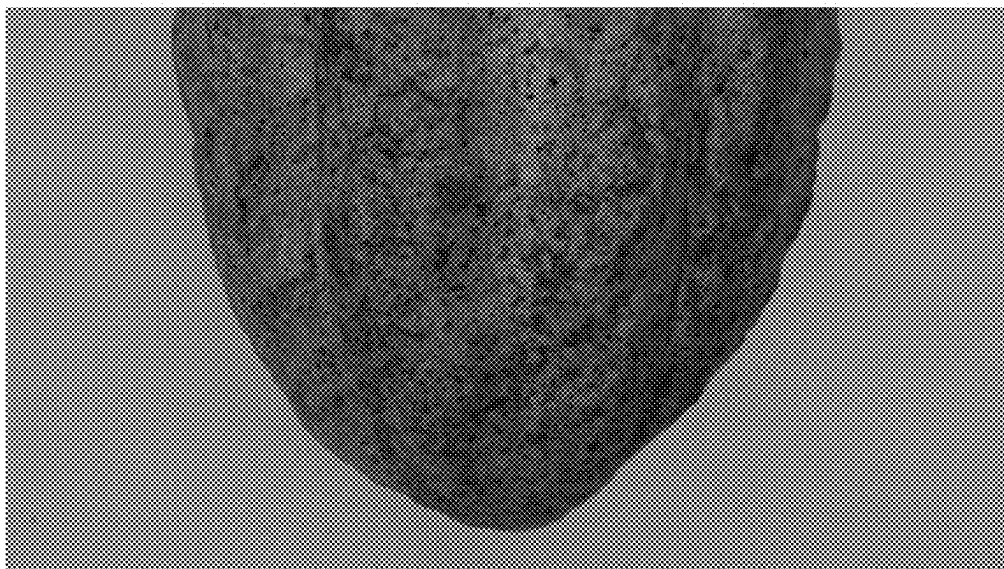


FIG. 3C



FIG. 3D

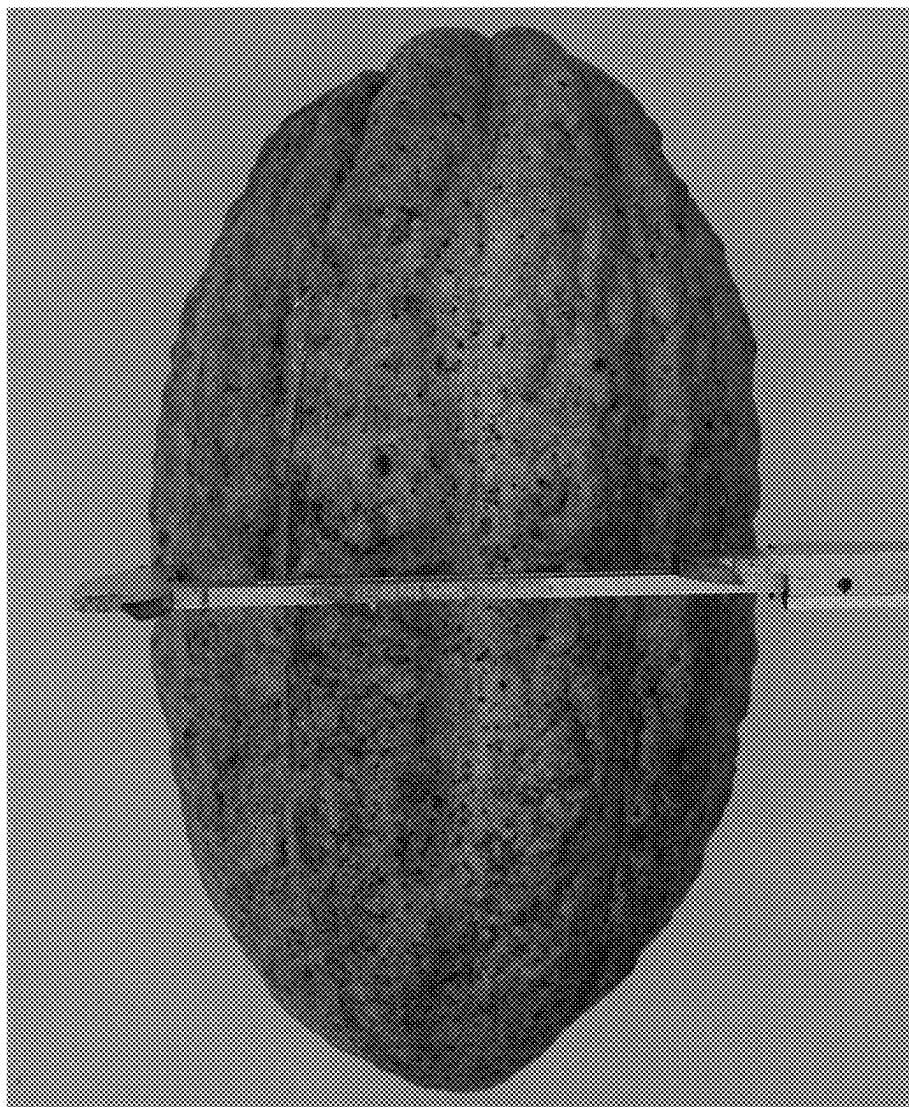


FIG. 3E

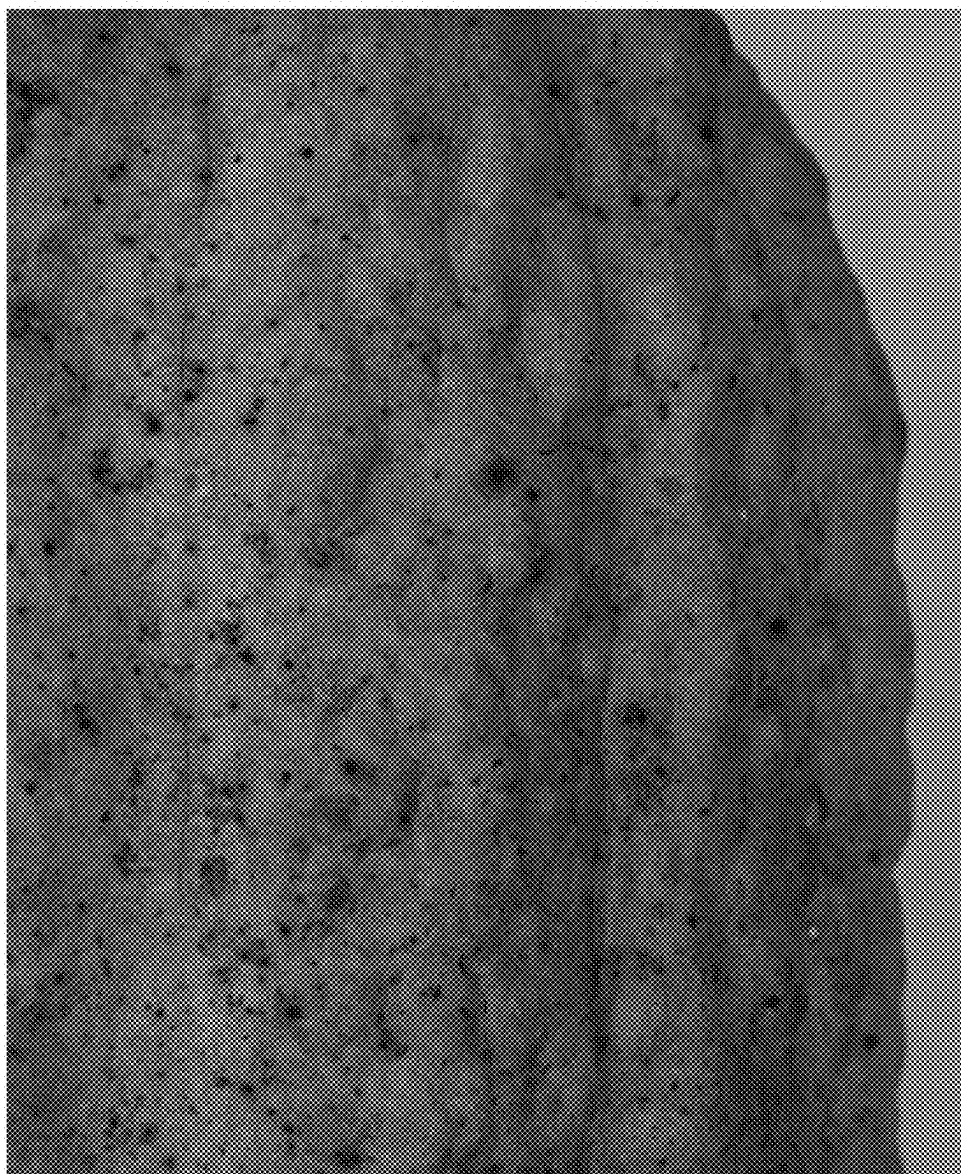


FIG. 3F

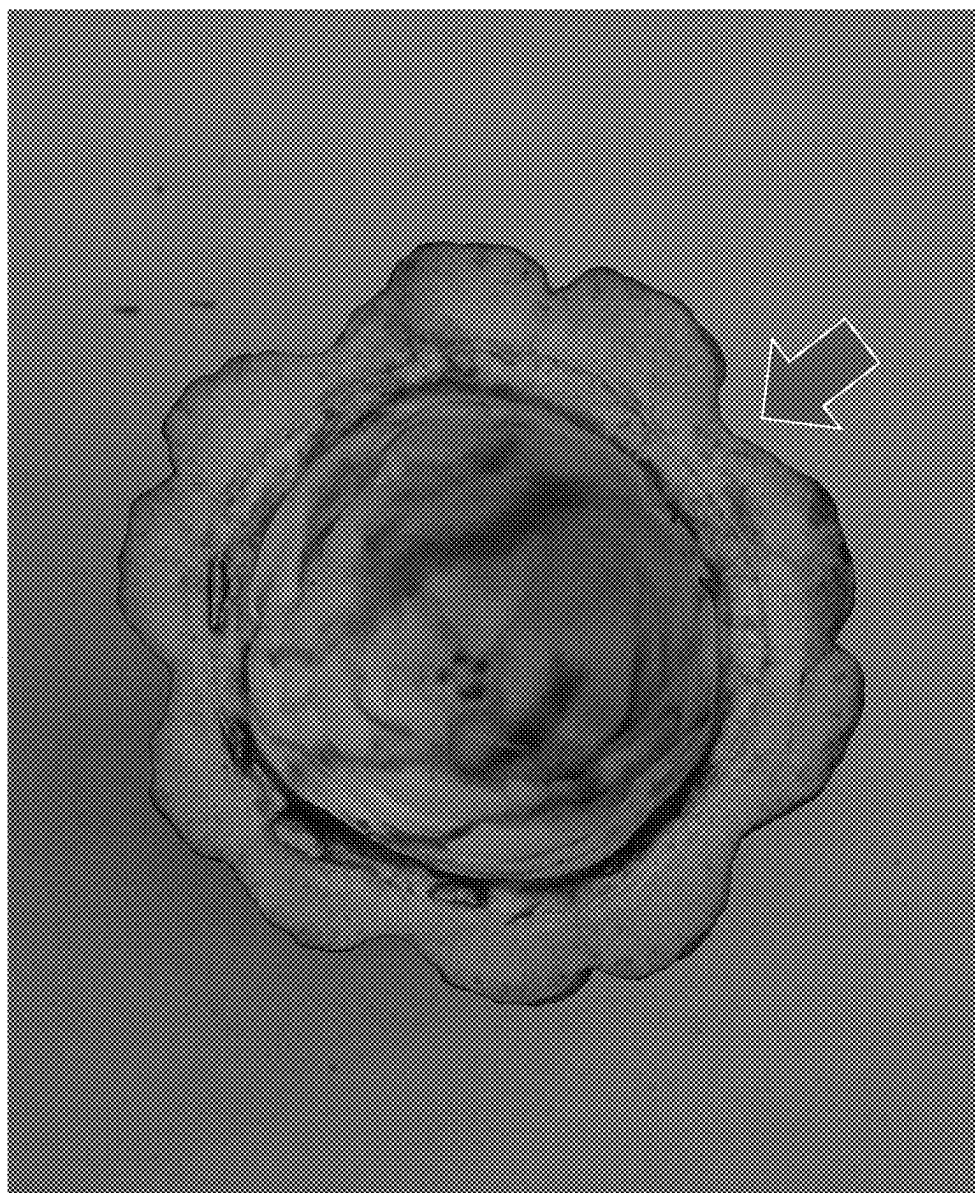


FIG. 3G



FIG. 3H

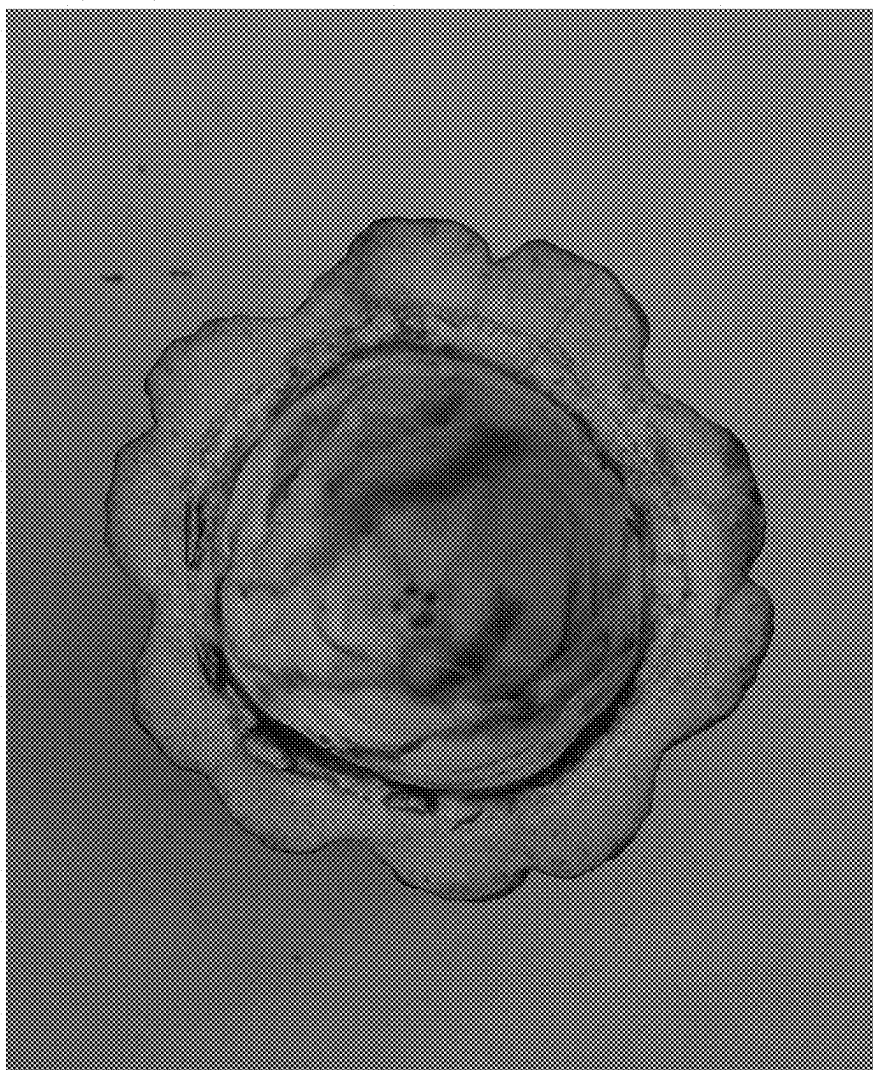


FIG. 3I

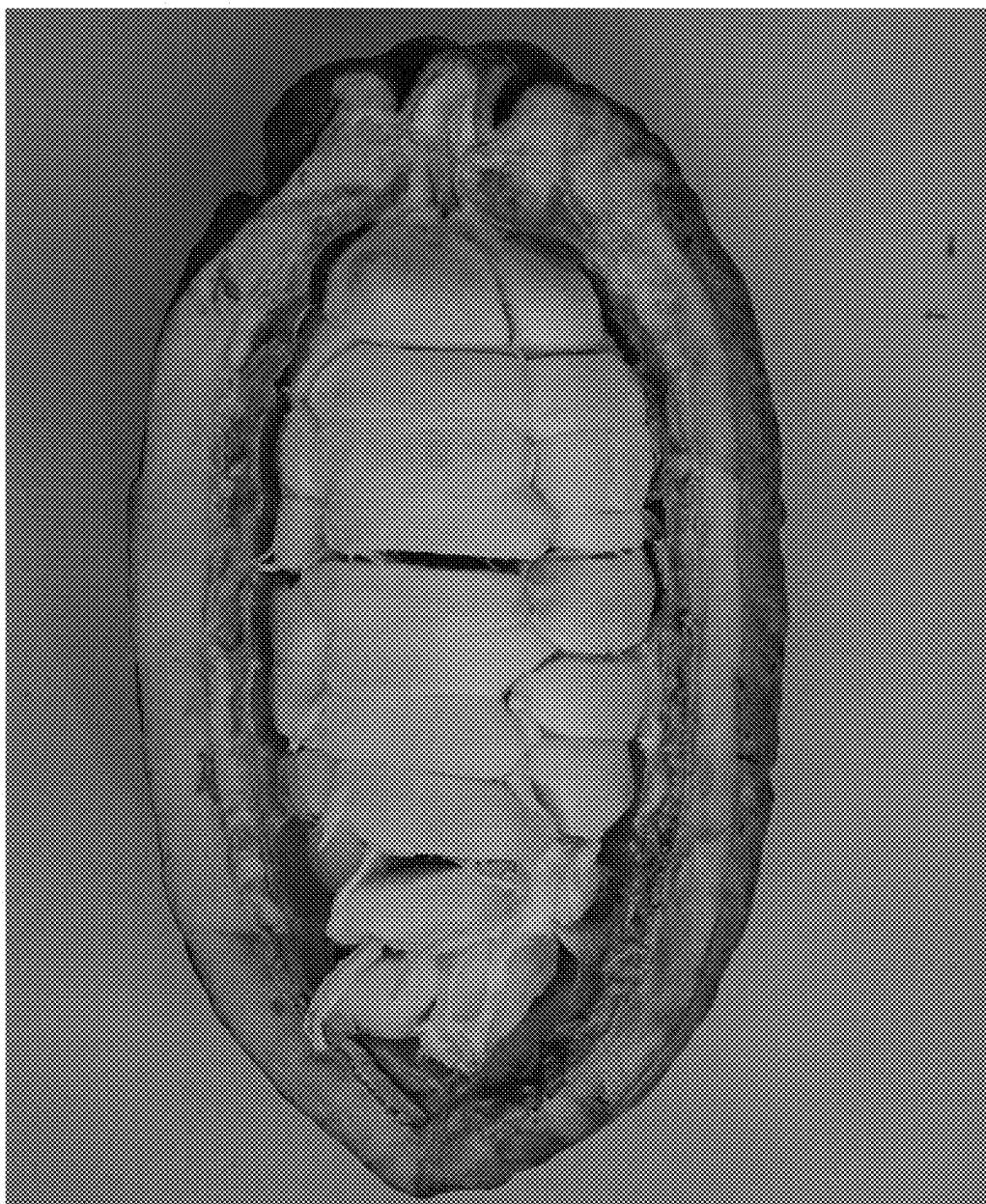


FIG. 3J



FIG. 4A

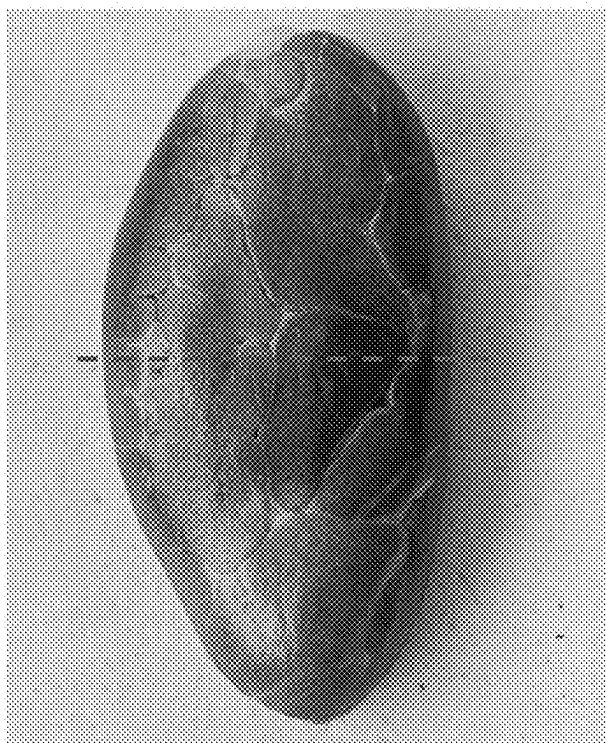


FIG. 4B



FIG. 4C

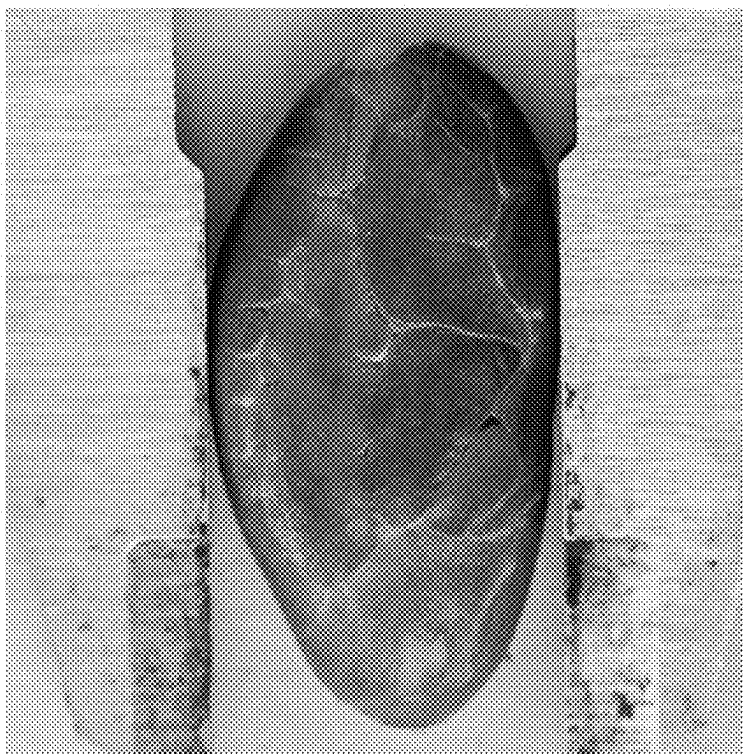


FIG. 4D

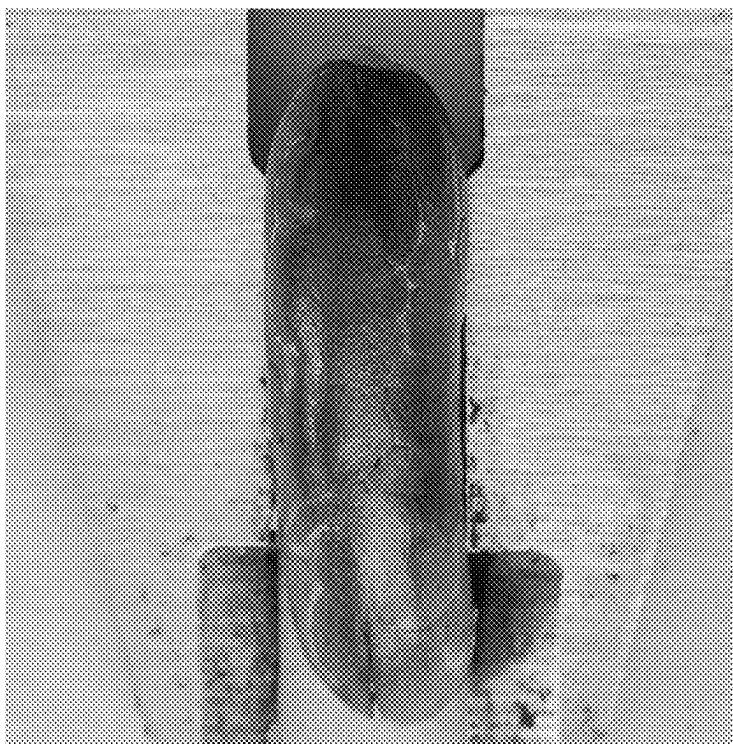


FIG. 4E

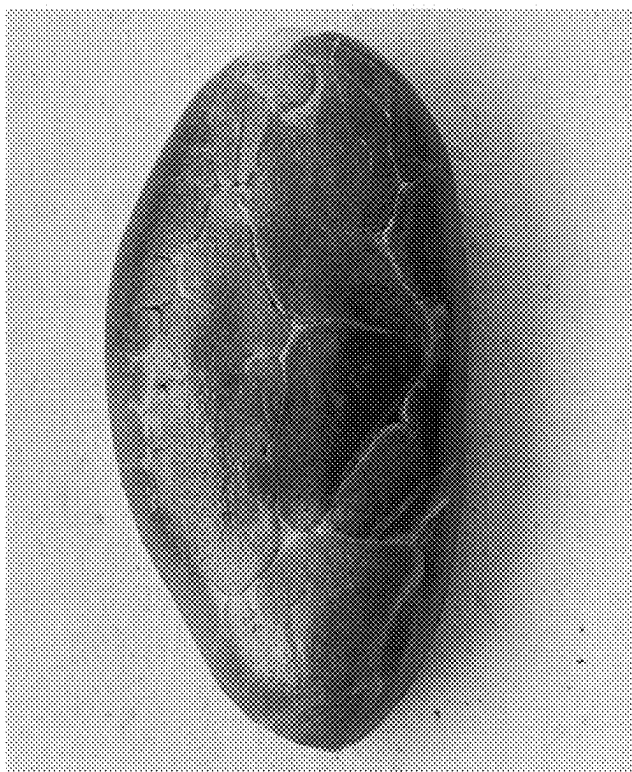


FIG. 4F

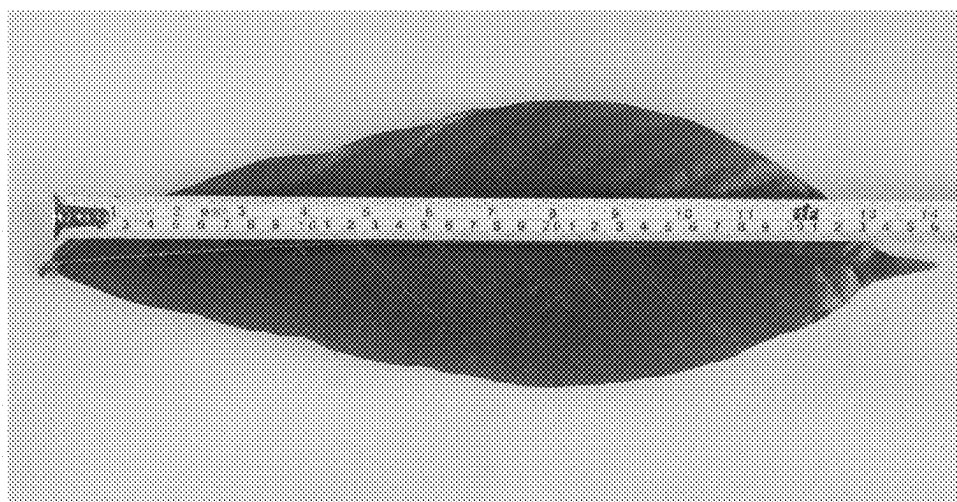


FIG. 5A

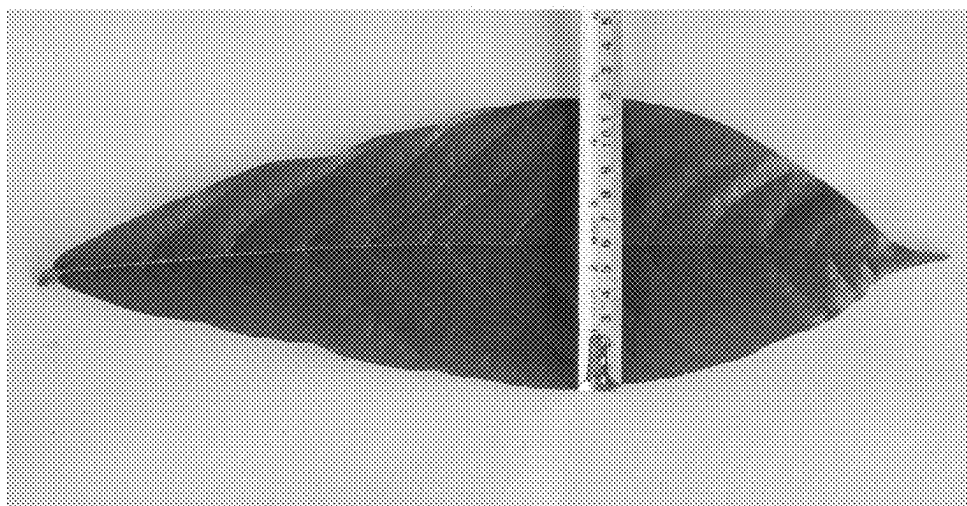


FIG. 5B

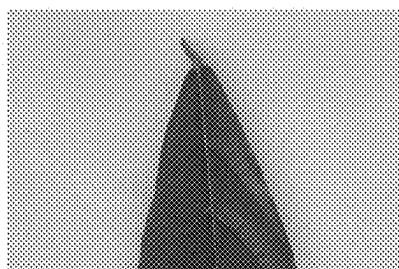


FIG. 5C

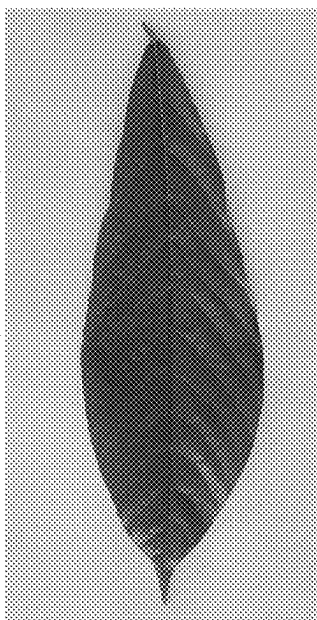


FIG. 5D

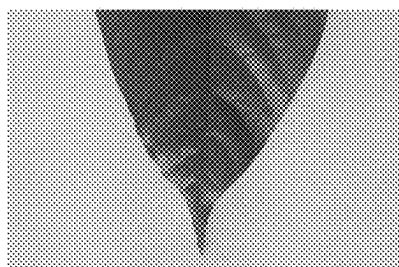


FIG. 5E

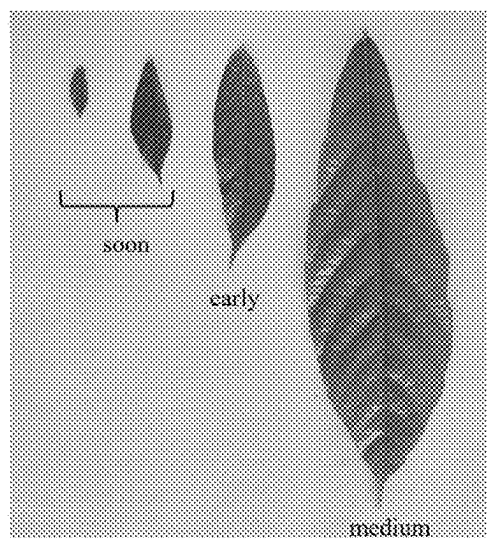


FIG. 5F

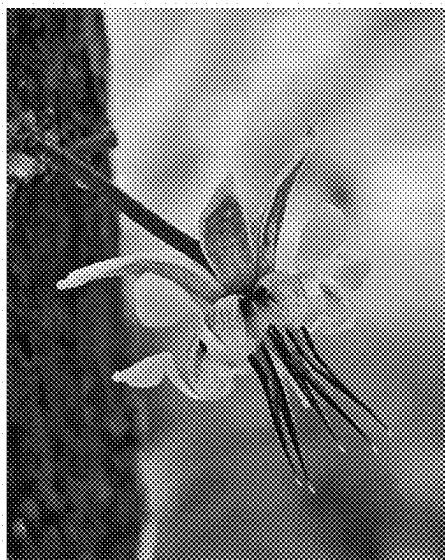


FIG. 6A



FIG. 6B



FIG. 6C

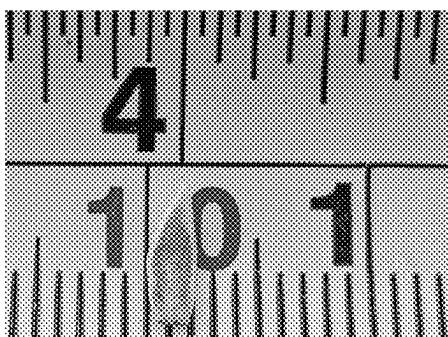


FIG. 6D



FIG. 6E

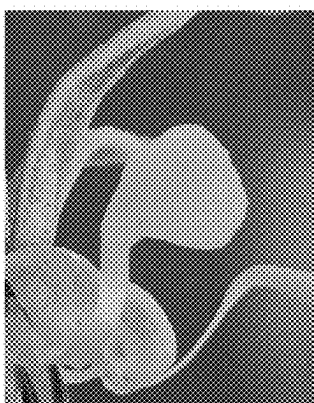


FIG. 6F

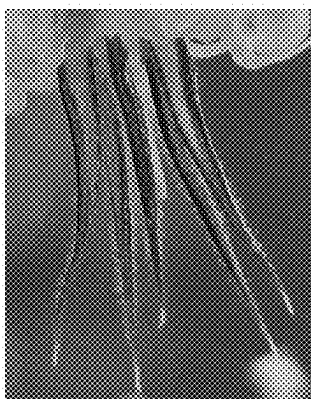


FIG. 6G

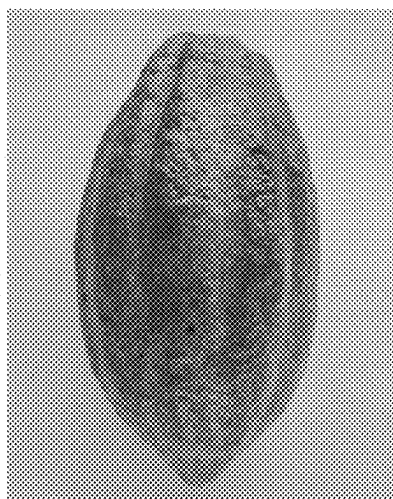


FIG. 7A

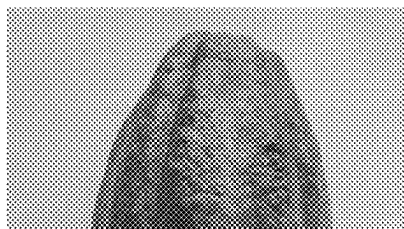


FIG. 7B

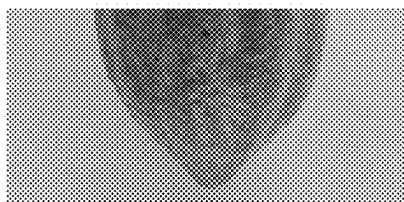


FIG. 7C



FIG. 7D

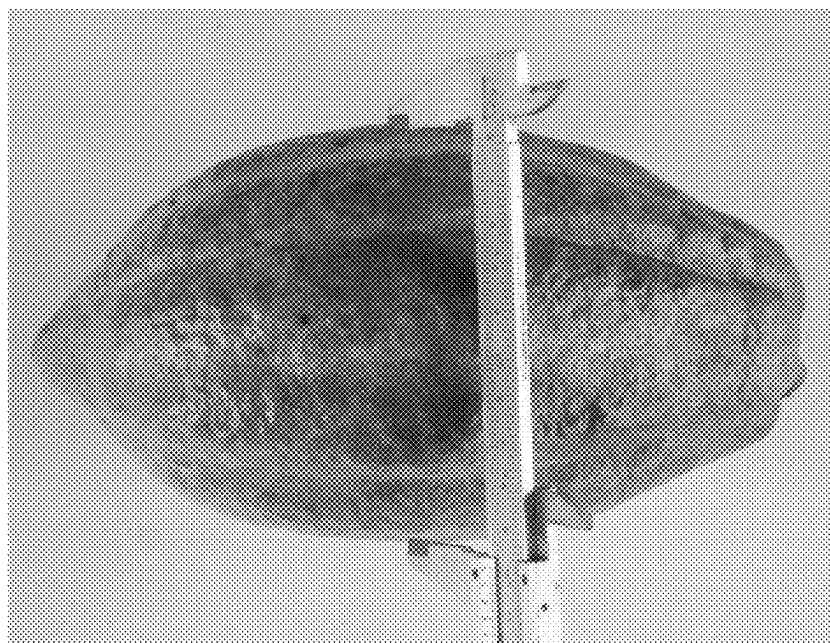


FIG. 7E



FIG. 7F

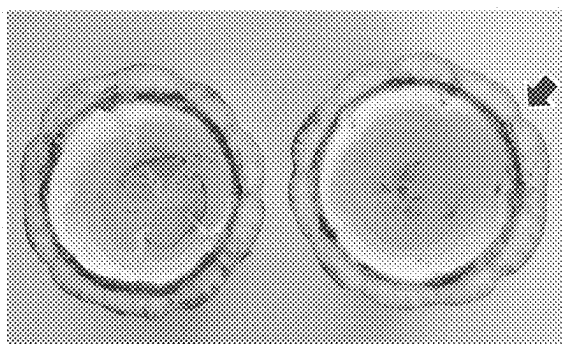


FIG. 7G

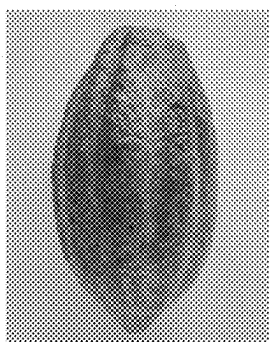


FIG. 7H

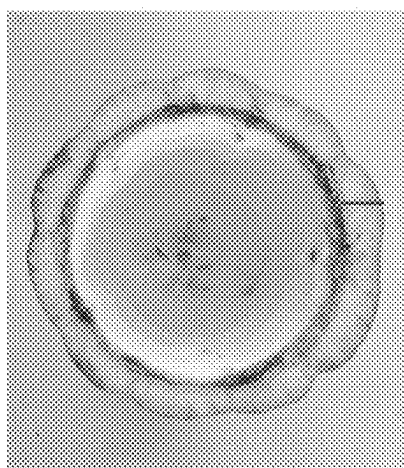


FIG. 7I

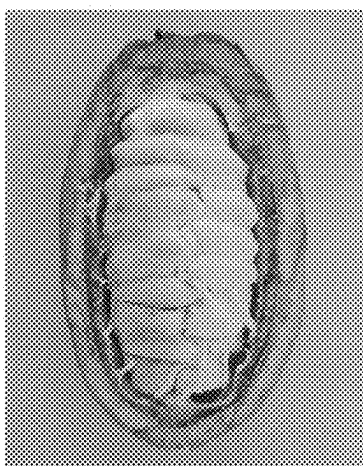


FIG. 7J

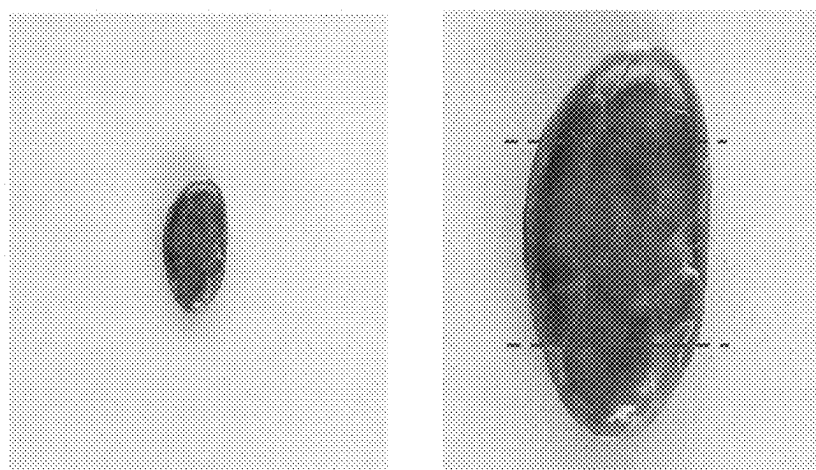


FIG. 8A

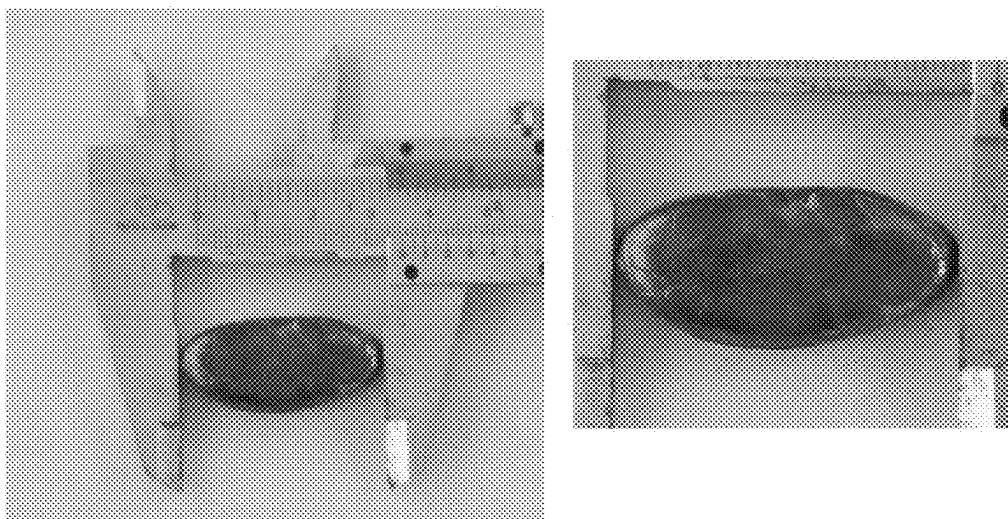


FIG. 8B

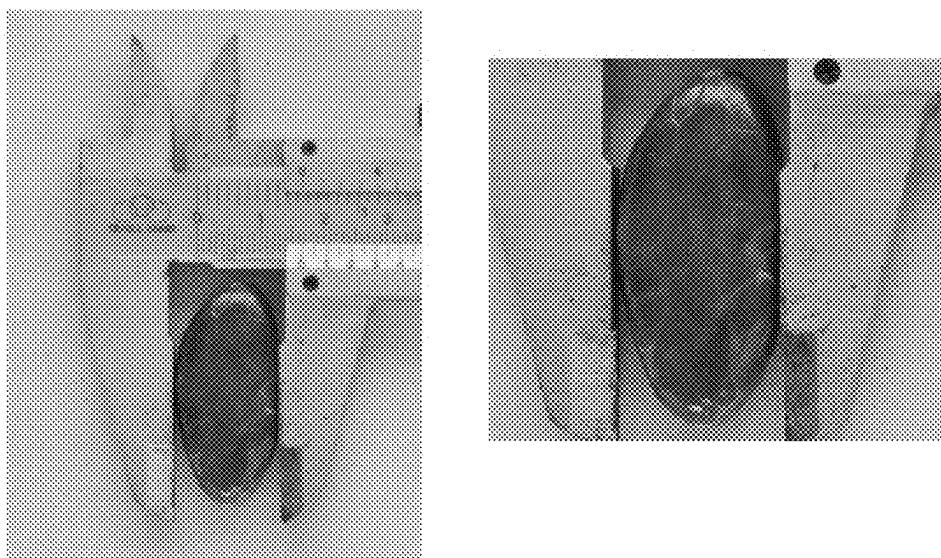


FIG. 8C

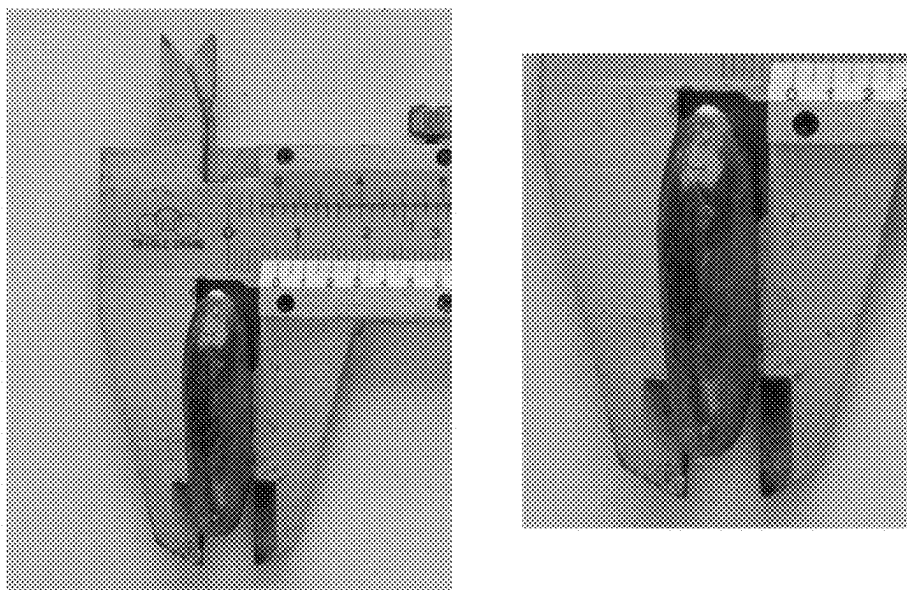
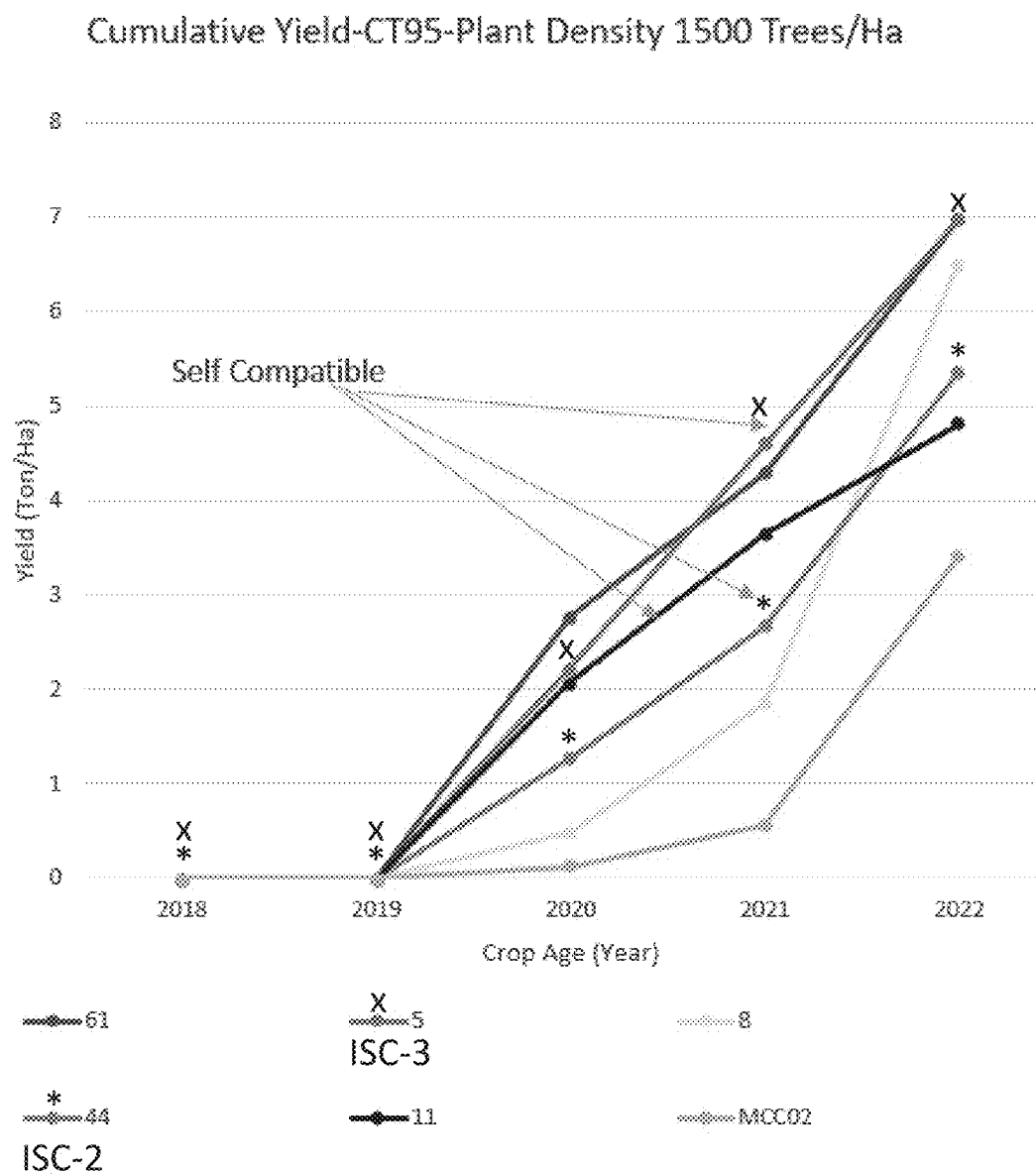


FIG. 8D



FIG. 8E

**FIG. 9A**

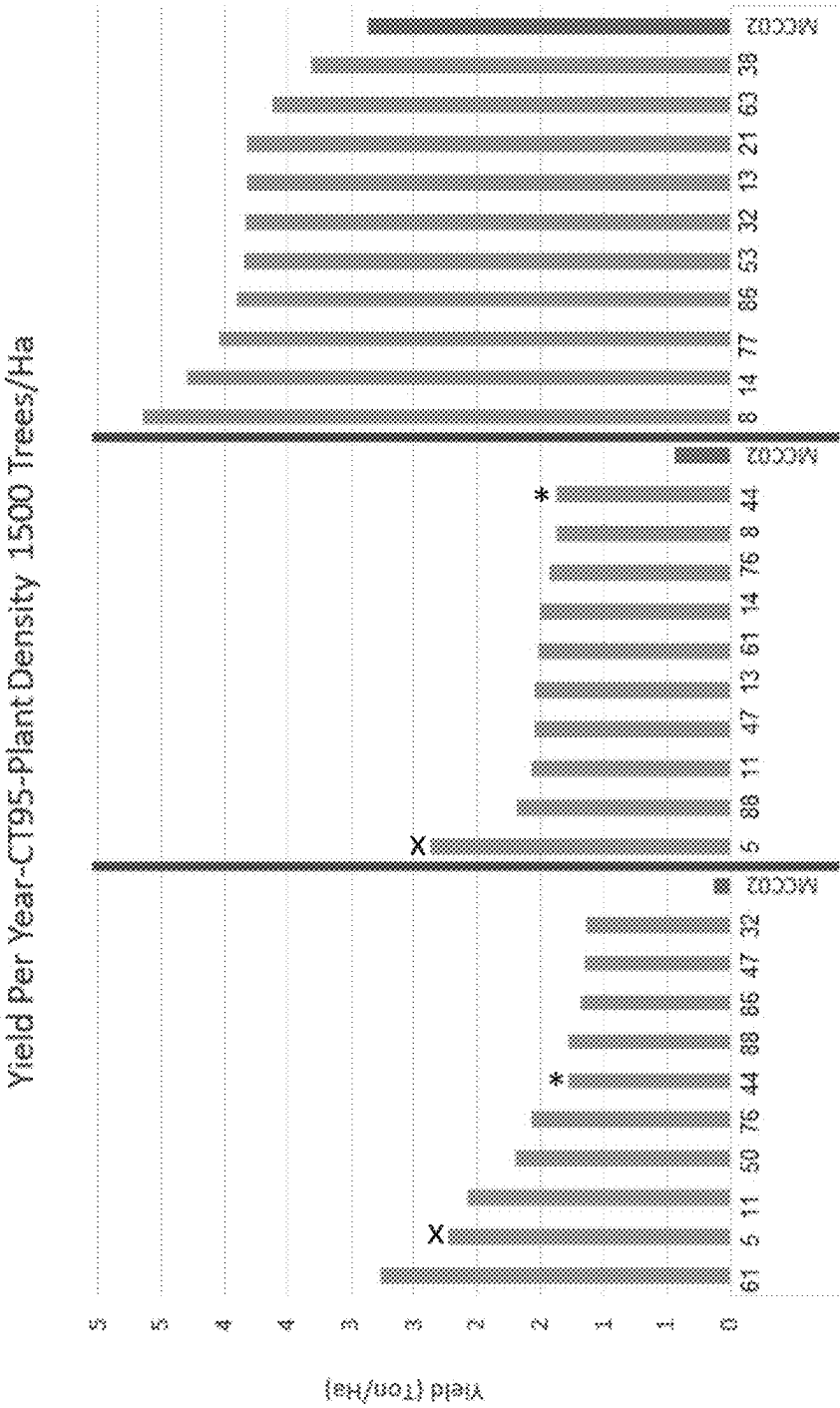


FIG. 9B

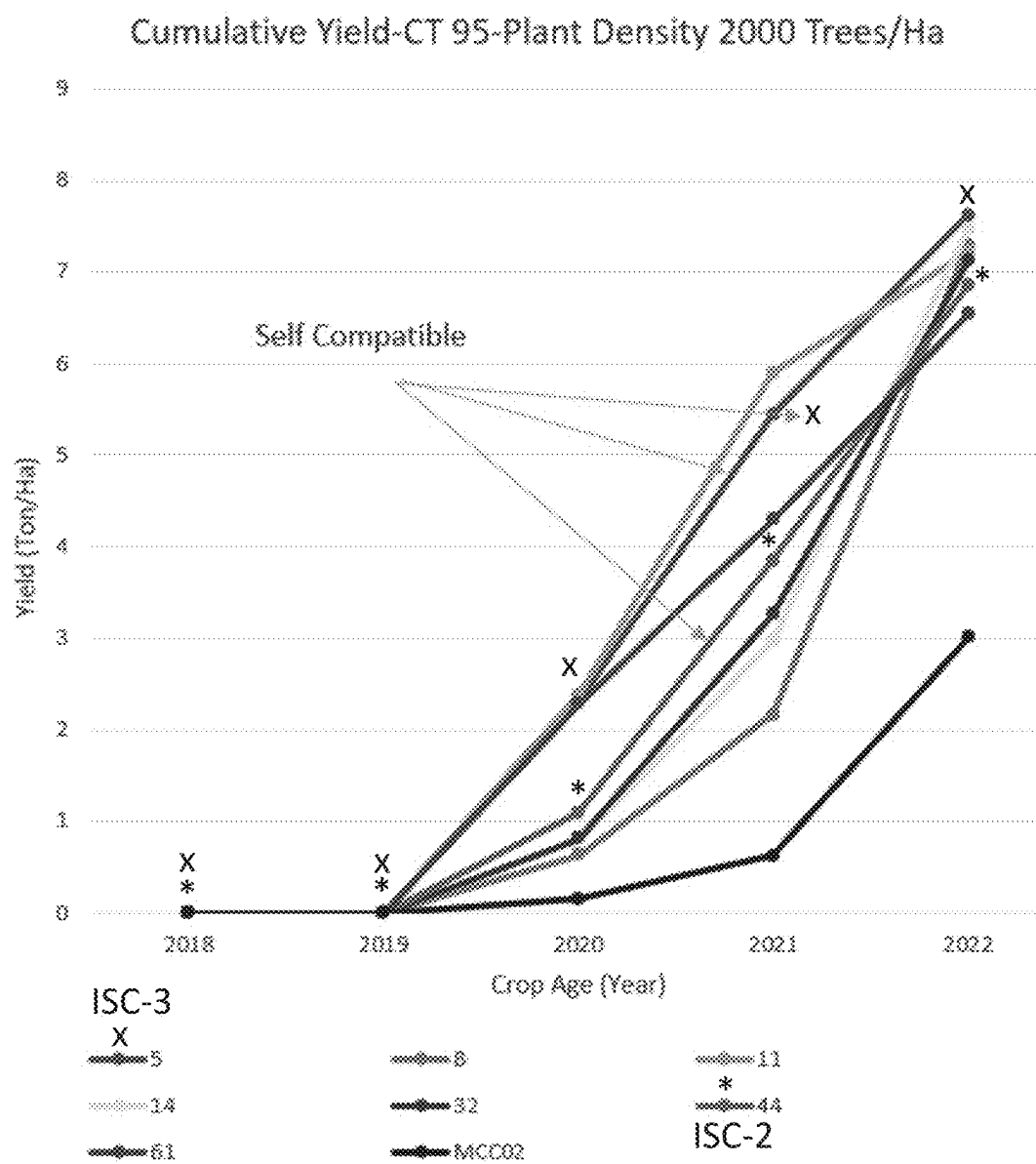


FIG. 9C

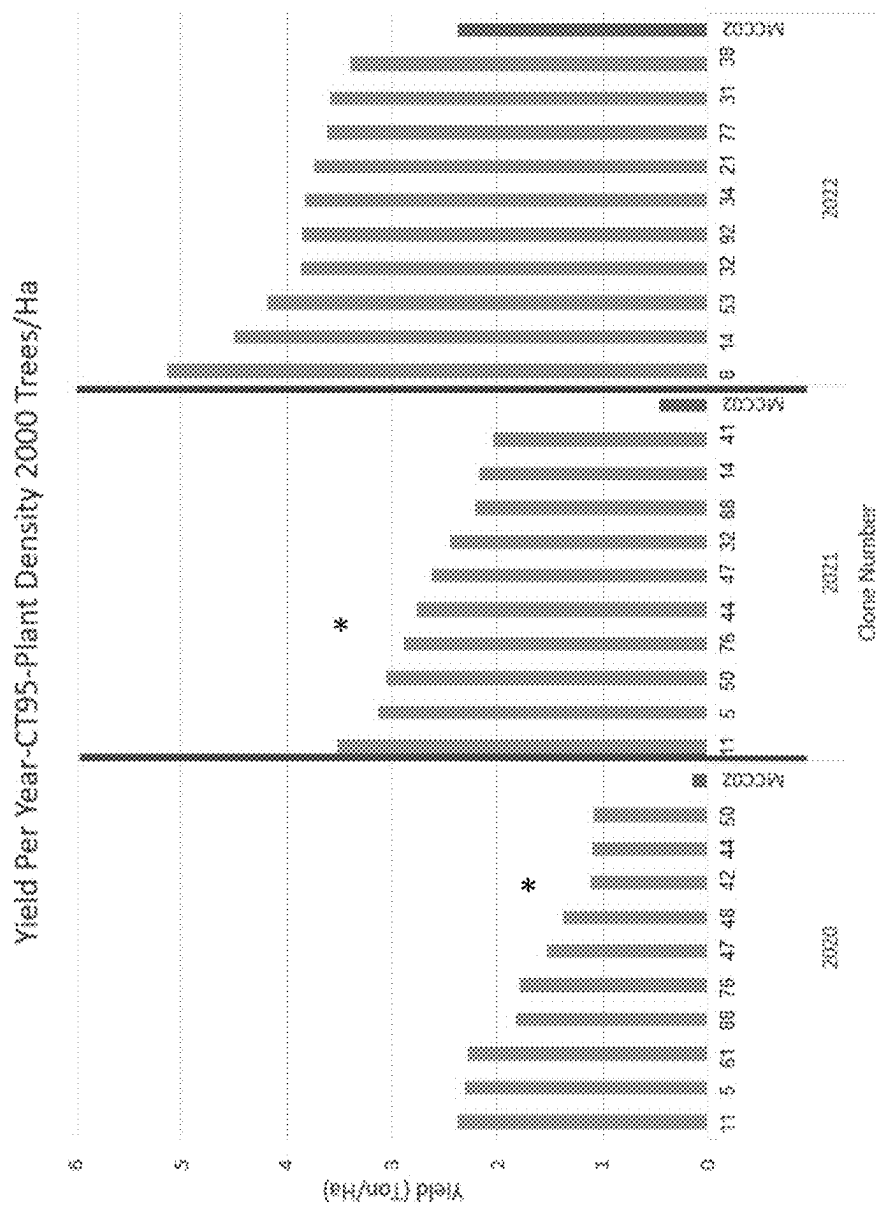


FIG. 9D

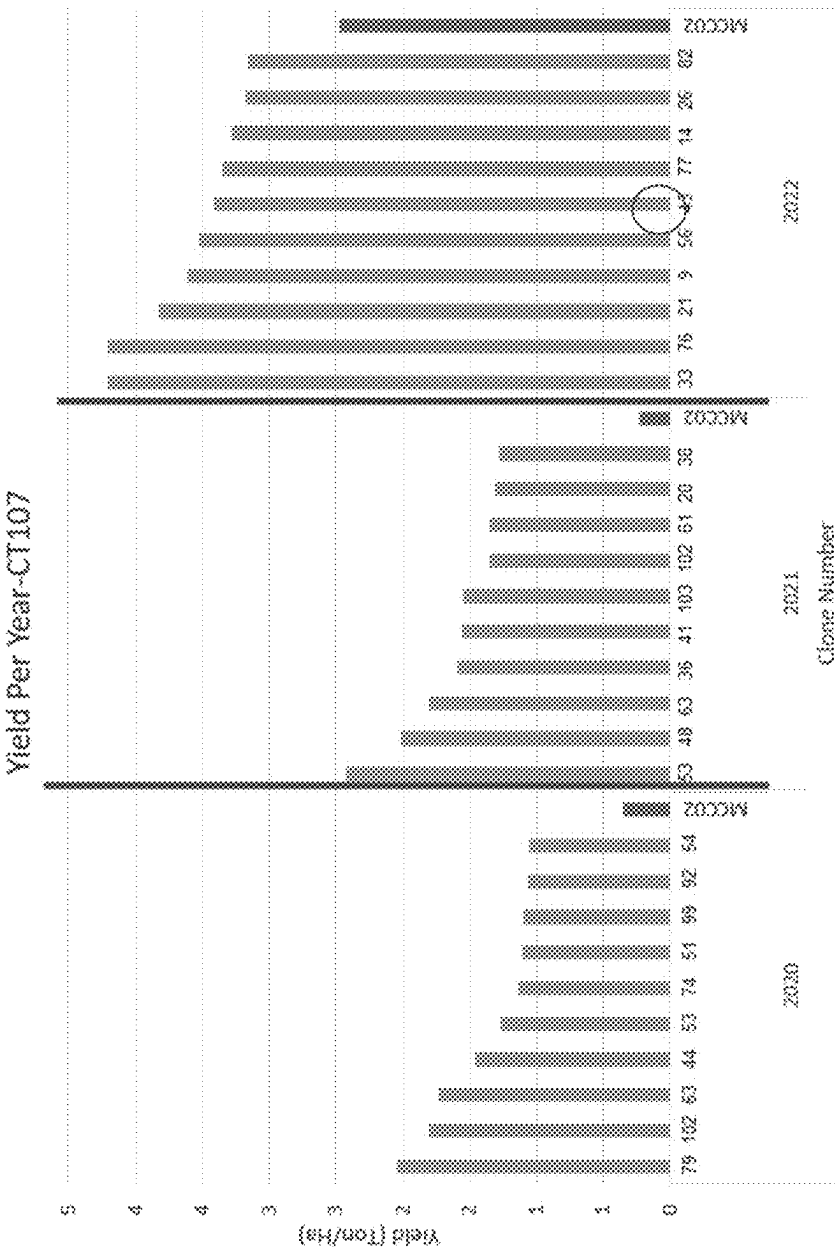


FIG. 9E

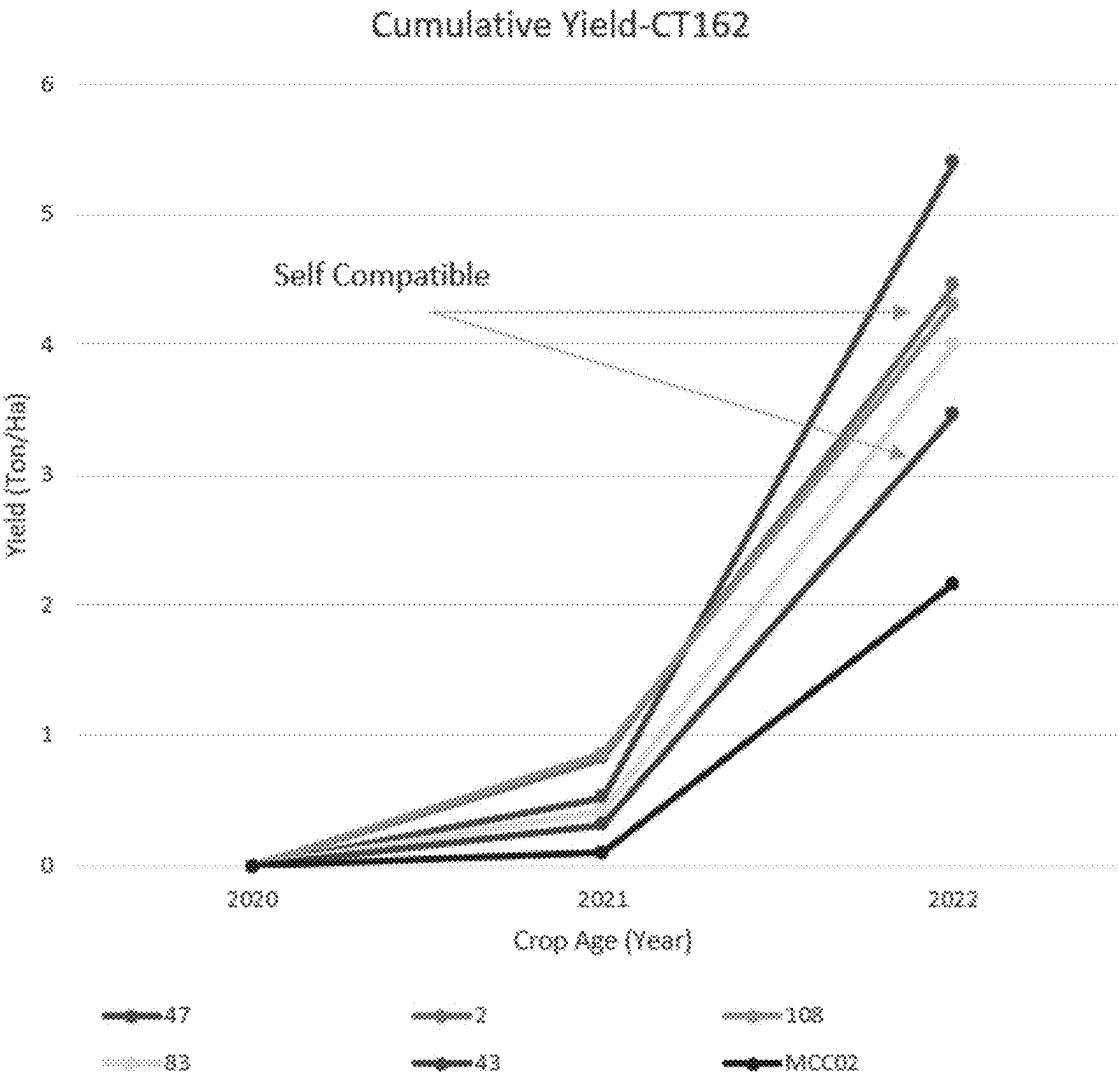


FIG. 9F

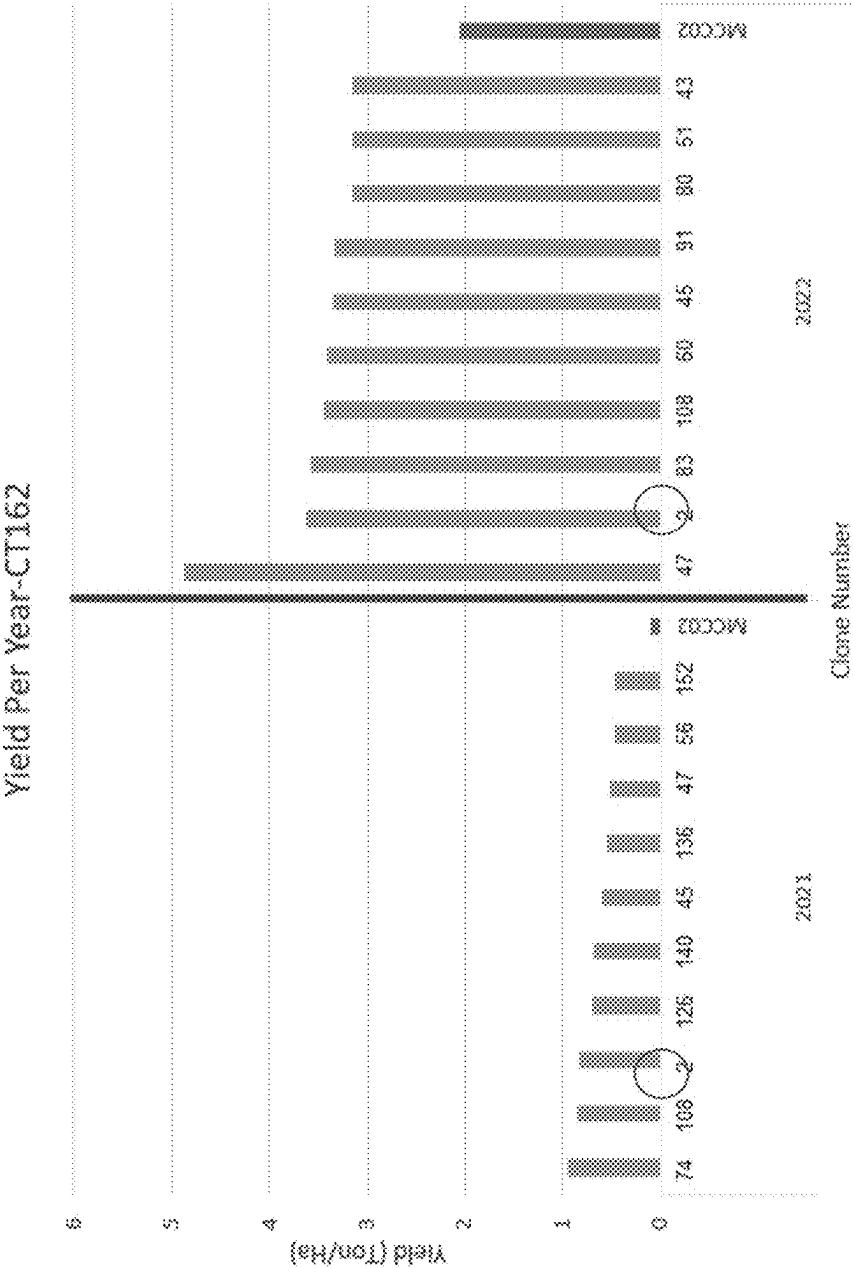
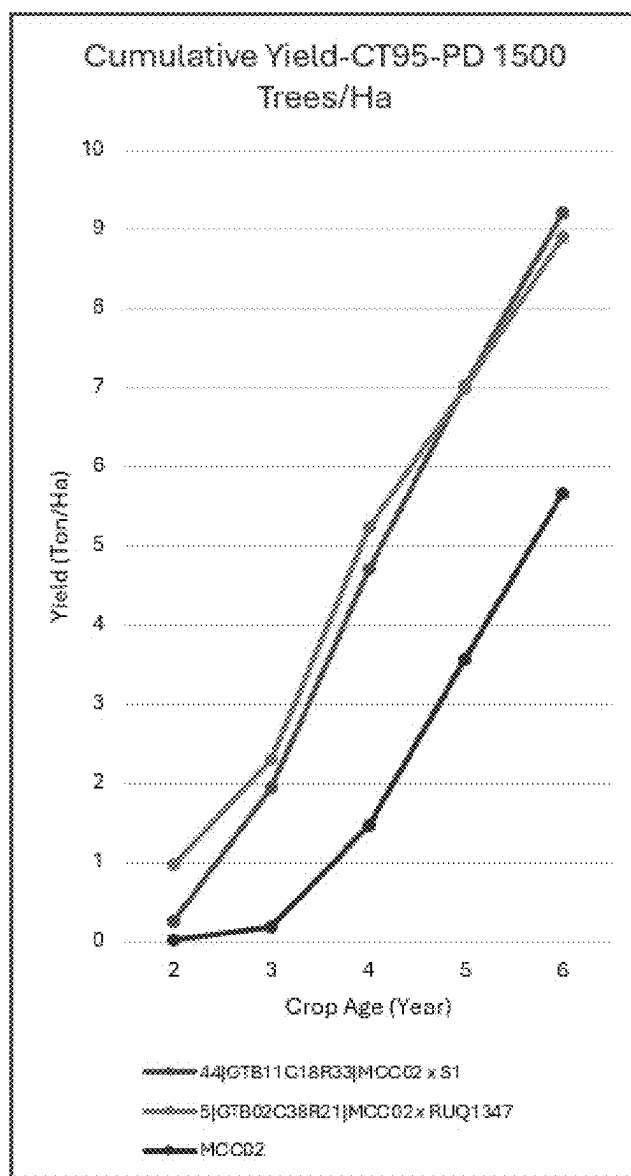
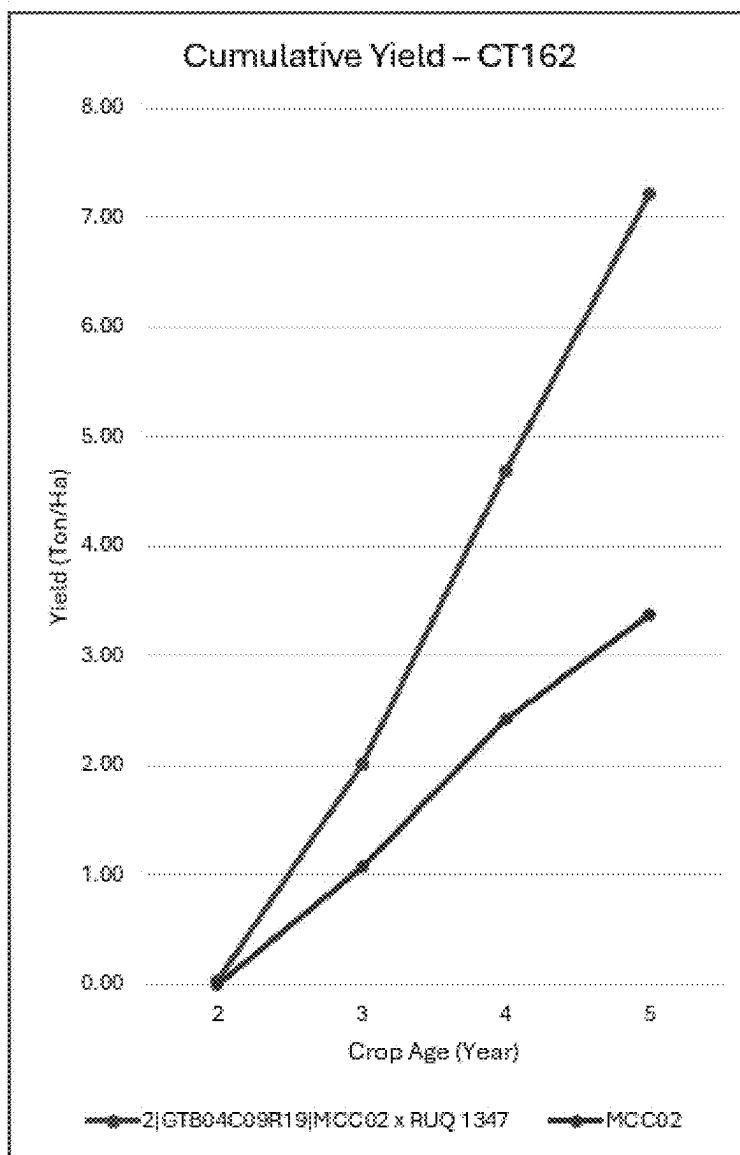
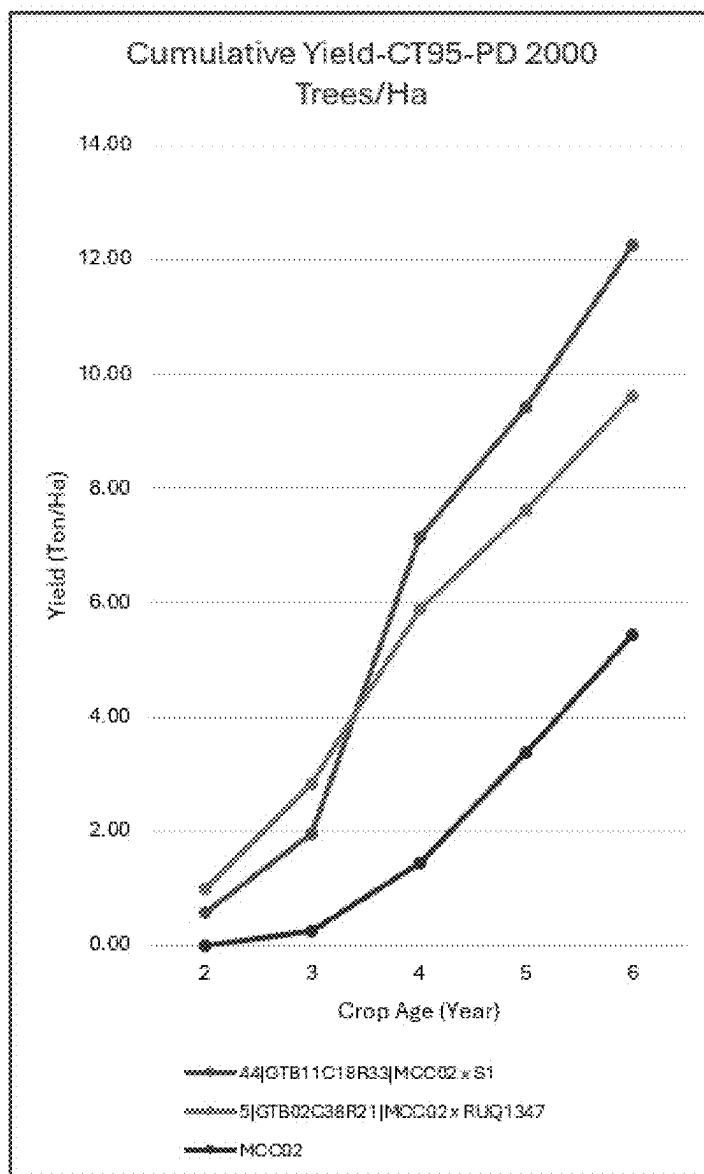


FIG. 9G

**FIG. 10A**

**FIG. 10B**

**FIG. 10C**

Progress Testing Self Pollination ISC-1

REP	Flowers Pollinated	Day 3	Day 5	Day 8	Day 15	Day 30	Final Rate	Avg
Rep 1 (Pkg)	20	15	11	8	3	3	15%	35%
Rep 2 (Pkg)	20	16	14	10	6	6	30%	
Rep 3 (Pkg)	20	18	18	16	12	12	60%	
Rep 4 (Pkg)	20	16	8	9	8	7	35%	45%
Rep 5 (Pkg)	20	14	14	13	13	11	55%	
Rep 6 (Pkg)	20	20	18	15	12	9	45%	
Grand Survival		83%	69%	57%	35%	35%	40%	StDev = 0,167

FIG. 11